

Technical & Business Writing

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(BS Physics)



Material Developed & Compiled by
Mubadirah Durrani

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PREFACE

The book “Technical and Business Writing” is an integrated guide for high intermediate to advanced learners at university. It focuses on development of the Physical paragraph (basic unit of composition) into forms of academic, business, professional and general purpose writing. The book is designed to be flexible enough to be used as a writing component in an intensive and independent writing course. This book is based on the concept that a student having basic skills of writing coherent paragraph can be metamorphosed into skilled writer in the field of Technical and Business Writing. The approach in each chapter is functional with model that can facilitate the students to comprehend with clarity, imitate the model to enhance their Technical and Business Writing skill so that they produce their own qualitative piece of writing confidently.

It is indeed a very pleasant book to use. As it fulfills the needs of the students, I am sure they will benefit from it.

Prof. Dr. Syed Zafar Ilyas

INTRODUCTION

Discussion with students and teachers at Physics Department at University brought the reality that students who lack knowledge and skill in the field of English language, do not lack intelligence. We have firm believe that such students can learn quickly and well if they are provided opportunity to improve their English Language Skills. This book is developed to give priority to writing skill in English language and enhance their capability in that.

The Purpose of the book “Technical and Business Writing” is to help students to improve command of English and get skill practice about expressing ideas with in a prescribed format in black and white. Another reason is to enable them to polish their potentials to create meaningful assignments and technical reports.

After going through various steps towards report writing, students deal with the rhetorical patterns found in expository writing. Academic outline format and brainstorming enable students precede clear and cohesive piece of writing.

Learning about the procedure of document design and several stages of writing forms, inculcate confidence in students to write research papers or thesis when they complete their research work.

It is an effort to increase the capacity of a student but students’ motivation and commitment generally what leads to success not only during their studies but also in their practical life.

CONTENTS

Unit 1:	Reporting and Gathering Information.....	1
Unit 2:	Technical Exposition	13
Unit 3:	Organising Information and Generation Solution.....	31
Unit 4:	Polisiing Style	47
Unit 5:	Document Design.....	57
Unit 6:	Citation.....	69
Unit 7:	Creating the Professional Report	75
Unit 8:	Reports	83
Unit 9:	Progress Report.....	93

Unit–1

REPORTING AND GATHERING INFORMATION

Written by: Mubadirah Durrani
Reviewed by: Syed Zafar Ilyas

OBJECTIVE

To write technical report and use library effectively.

1.1 Technical Report Writing

A report is formally written account which is governed by terms of reference known to both writers and readers. It is carefully compiled from data available to the writer and directed to wards achieving a specific purpose.

- What are reports for?
- To record routine events
- To record non-routine events
- As the bases for making decisions
- As a bases for avoiding decisions

There are three purposes for communicating information in reports.

- To describe
- To record
- To persuade

Once you know why you are writing the report, what type of report it should be, how much relevant material you have, you will be able to proceed.

Ask yourself:

- What are the general aims of the report (to inform, to persuade, to place on record, to provide a bases for action)
- What is the subject of the report (a visit, an inspection an accident)
- How much information is required (essential facts only, a detailed analysis, a factual record, mainly recommendations with the necessary supporting evidence).
- What form of report is required (memo, letter, short, long, informal, formal)
- Who will read the report
- Who will get the final report

After determining the terms of reference and a precise picture of whom you are writing for, you are in a better position to set yourself objectives. These should be stated as clearly, concisely and accurately as possible. The objectives are a refinement and a closer definition of key points.

1.1.1 Gathering Information

Now it is time to collect the required information. Information can be collected in many ways. You may do it by interviewing people, by reading or by research, by carrying out tests, or by observation etc. it is better to collect more information than you think you will need. It is easier to write when you are able to choose from more than you need. When you are researching your report, it is better to bear in mind the following questions:

Who? What? Where? How? Why?

During the assessment stage see that your preparation is complete.

- Have all sources of information been consulted?
- Have all relevant aspects of the subject been covered?
- Is the preparation according to the terms of reference, readership analysis and objectives?

During the planning stage you will need to:

- Decide on the type of the report
- Select the information
- Plan arrangement of the selected information
- Organise the body of the report.

1.1.2 Arrangement of Information:

Many short, informal reports will comprise the following four elements:

- Summary
- Introduction
- Body
- Conclusions and / or recommendations

The arrangement best suited to your report can be chosen. Many variations are possible. It does not particularly matter what the arrangement is, so long as it is there. It needs to be clear and logical to the reader.

1.1.3 Summary:

The terms ‘synopsis’ and ‘abstract’ tend to be used in more academic and scientific documents, while summary is the most generally used of the three expressions.

Summary is provided for those readers who are in a hurry and still need an overall picture. They want the quick version so good summary should be ensured. Summarize the entire report body – introduction, discussion conclusion and recommendations.

1.1.4 Introduction:

This should inform the reader about why the report was commissioned, what circumstances led people to believe that a report was needed what was the purpose of the report and what issues the report covered. This will serve to provide background information about the subject before the body of the report. A good report should contain, in its’ introductory paragraph, a clear exposition (lay out in detail) of the theme of the report. It should contain in miniature the materials to be developed in the rest of the report. Subsequent paragraph should then fulfill the expectations set up in the readers mind by the introduction.

1.1.5 The Body:

Gives an account of how the conclusions and recommendations were arrived at. The development of your exposition throughout the paragraphs should offer the reader a smooth logical transition from start to finish. The final section should not come as a surprise.

1.1.6 Conclusion and / or Recommendations:

An overview is provided – a summary of all that has gone before. A specific context for the recommendations you are about to make is given.

Recommendations concern specific actions that should flow from the conclusions drawn. Keep in mind that just as recommendations are based on conclusions, so too are conclusions based on what you have discussed in the body of the report. No new material should be introduced in the conclusions or recommendations.

One of the skills which you should learn if you want your report to hold together well is the ability to be cohesive in your writing. Cohesion refers to the property

of a text where all part as logically related to form a unified whole. In a cohesive piece of writing the arrangement or linking of ideas comes across as a collection of loosely related statements and the message is not confused.

1.1.7 Features of a Cohesive Report:

- Consistent tone throughout.
- Unity of ideas within each paragraph. (Refer to the paragraph writing).

The concluding paragraph should settle what was raised in the first paragraph.

- Use of linking devices. Linking devices can be classified as follows:
 - a) The words ‘this’, ‘these’, ‘that’, and ‘those’ either as demonstrative adjectives or as pronouns standing for nouns. These words refer back to topics in previous sentences or paragraphs, they signal links with other topics being introduced.
 - b) Synonyms. Different ways of saying the same things, synonyms help to avoid monotony of expression while linking different aspects of same topic. Care should be taken to retain the same tone.
 - c) Generalisations. These contribute to economy of expression.
 - d) Connectives used to combine ideas according to the mood in long sentences.

So much for the structure of the report, let’s now consider the technique of outlining. An outline is not only an important planning tool but also exerts a strong influence upon the detailed structure and layout of the report, particularly the body section.

If you have only a rough outline or plan, and decide to jump into the writing process you may regret such action. An outline is a spatial or graphic tool for creating structure. It provides a writer with an hierarchy of headings and various levels of sub-headings under which ideas can be organized according to importance. As such an outline can help you to quickly identify any material you have omitted. An outline can also help you to determine the ideal sequence of ideas in your argument, and can be used to check whether you are giving too much, not enough or appropriate emphasis to any given part of your argument.

It is desirable to write a report within as short a period of time as possible. As you write, do not worry too much about style or the quality of your writing. Concentrate on getting the information on paper as quickly as possible. You will have the chance to make as many changes as you wish when you come to the review stage. That is when you can amend, delete and add things in order to polish your writing so that it communicates efficiently and effectively.

The body of the report should be written first. The introduction should be written next.

Keep the following principles in view while writing a report:

A report must show by whom it was made and to whom it is addressed.

It usually has a title or heading and the date.

Be certain of the aim(s) of your report and the subject(s) you are supposed to deal with.

Give all the information and put it in the right order.

The facts should follow a definite order, often a time order or sequence. Do not involve yourself in long descriptions / narratives or arguments. The facts should be stated clearly and concisely.

Try to recall the situation or event as best you can and give figures where necessary. Also give details of the time, date and place of any event to which it refers. A report has to convey information to the person who is reading it.

You can use sub-headings and tabulate your work so that the information is clearly set out and easy to find. The sub-headings should be arranged in a logical order.

Paragraphs can be numbered in a long report.

1.2 HOW TO USE THE LIBRARY

After checking your library's general reference books, you should find out whether the library has any specific books that would give you helpful material on the subject you are researching.

Every book that the library owns is listed on an index card. These index cards are

assembled in the *card catalog*. (The librarian will tell you whether the book is on the shelves or has been borrowed.)

How can the card catalog tell you where to look for the book on the shelves?

On page 12 are examples of catalog cards for the same book. Each important item of information is labeled on the author card and explained below.

The *call number* is printed in the upper left-hand corner of card. This number is part of a system which outlines all human knowledge. There are two systems in general use: the Dewey decimal system and Library of Congress system. Each is a means of organizing books so that new title can be easily and regularly added without changing the system in any way. Because the Dewey decimal system is more common.

1.2.1 The Dewey Decimal System: This system of organization divides library books (except works of fiction and biography) into ten classes with numbers which identify them on the shelves.

000-099	General works (reference books, encyclopedias, bound periodicals, almanacs)
100-199	Philosophy, psychology, conduct
200-299	Religion, mythology
300-399	Social sciences, economics, government, law
400-499	Language (includes dictionaries, grammars)
500-599	Natural sciences (mathematics, biology, etc.)
600-699	Useful arts (includes agriculture, engineering)
700-799	fine arts (includes sculpture, painting, music)
800-899	Literature (includes poetry, plays, essays)
900-999	History (includes geography, travel)

Each of these classes has subdivisions. For example, literature is divided into:

800-809	General
810-819	American literature
820-829	English literature
830-839	German literature
840-849	French literature
850-859	Italian literature
860-869	Spanish literature

There are additional subdivisions, so that under literature we have special numbers designating poetry, drama, fiction, etc. By the use of decimals, the numbers are broken down further. Early English Drama 1066-1400, for instance is 822.1. These numbers are called the class numbers. They are combined with the initial of the author's last name to give the *call number*, by which a book is classified.

Some libraries use a system of their own or the Library of Congress system. Whichever system is used, the most important key to the location of any book in a library is the call number.

Small libraries keep their card catalogs as simple as possible. Some libraries have an *Author Card*, a *title card*, and a *Subject Card* for each book they own. In the upper left-hand corner of these cards is the call number.

Author Card: if you know the author's name you can locate the title of any of his books, listed alphabetically, that are in the library.

Title Card: If you do not know the author's name, but you know the title of the book, you can find both title and author on the title card. Title cards are placed in alphabetical order according to the first word of the title of the book, except for the articles *A*, *An*, or *the*: *Last Days of Pompeii*, *the*.

Subject Card: Though many libraries do not cross-refer the cards in their catalogs under general headings, some do.

Cross-reference Card: If the topic you are researching is limited, you may not easily find specific books that would help you. In that case look in related fields. Cards which say "see" or "see also" will guide you in directions that may prove fruitful.

The Magazine File: After researching the card catalog, look next into the *Reader's Guide to Periodical Literature* which lists magazine articles. Are there any articles that might shed light on your topic? Check with the librarian to learn whether the magazine containing those articles are available. Make a careful record of unavailable magazines because you might find them in another library or at home, or be able to borrow them.

Newspaper Index: *The New York times Index*, available in many libraries that subscribe to the paper, offers a complete index to articles that have appeared in *The New York times*.

Exercise: Answer the following questions by referring to the library card below.

274.7	Greek Church in Russia.
H44r	HECKER, JULIUS FRIDRICH, 1881-
	Religion under the soviets, by Julius F. Hecker. New York, Vanguard Press, 1927.
	xvii, 207 p. 18 ¹ / ₂ cm (Vanguard studies of Soviet Russia) "Selected Bibliography": p. 204-207.
1.	Russia - Religion.
2.	Greek Church in Russia. I. Title
	27 - 23706
Library of Congress	BR936 .H3
Copy 2.	
Copyright A 1010219	28d3

1. What is the title of this book?
2. Who is the author of this book?
3. When was the author born?.....
4. Who is the publisher of this book?
5. What is the date of its publication?.....
6. What is the call number?.....
7. What are the suggested subjects under which this book could be found in the card catalog?
8. Are there any illustrations, maps, tables or diagrams in this book?
9. What is the total number of pages in this book?

1.2.2

Indexes (cont.)

Book Review Digest
Business Periodicals Index
Education Index
International Index to Periodicals
New York Times Index
Readers' Guide to Periodical Literature
Social Sciences and Humanities Index

Encyclopedias

New Catholic Encyclopedia
Compton's Pictured Encyclopedia
Encyclopedia Britannica
Encyclopedia Americana
Encyclopedia International
The Jewish Encyclopedia
Encyclopedia of Islam
The World Book

Unabridged Dictionaries

A Dictionary of Americanisms on Historical Principles
Oxford English Dictionary
Webster's New International Dictionary
The Random House Dictionary of the English Language
Funk and Wagnalls New Standard Dictionary of the English Language

Biographical Dictionaries

Current Biography Dictionary of American Biography
Dictionary of National Biography
Who's Who
Who Was Who

Yearbooks and Annuals

Statesman's Year-Book
World Almanac and Book of Facts
Information Please Almanac
Yearbook of the United Nations

Atlases and Maps

Atlas of Islamic History
Collier's World Atlas and Gazetteer
Encyclopedia Britannica World Atlas
Hammond's Geographical Atlas
The Times Atlas of the World

Books of Quotations

JOHN BARTLETT, Familiar Quotations
Hoyt's New Cyclopedia of Practical Quotations
S. G. CHAMPION, Racial Proverbs
The Oxford Dictionary of Quotations
BURTON STEVENSON, The Home Book of Quotations

1.2.3 Specific Reference Works

Business and Finance (cont.)

ROBERT J. SCHWARTZ, Dictionary of Business and Industry
UNITED NATIONS, Statistical Yearbook
Foreign Commerce Yearbook

Education

American Universities and Colleges
CARTER AND BURKE, How to Locate Educational Information
Education Abstracts
Encyclopedia of Educational Research
Review of Educational Research

Engineering

LOUIS DERR, Cyclopedica of Engineering
A. L. DYKE, Automobile and Gasoline Engine Encyclopedia
Engineering Index
Industrial Arts Index

Humanities

American Authors, 1600-1900
J. BLANCE, Bibliography of American Literature
Cambridge Bibliography of English Literature
Cambridge History of American Literature
Collins' Music Encyclopedia
Cyclopedia of Painters and Painting
Encyclopedia of World Art
Encyclopedia of Religion and Ethics
Grove's Dictionary of Music and Musicians

Humanities (cont.)

Twentieth Century Authors
Cambridge History of English Literature
EGON WELLESZ, Ancient and Oriental Music

Natural Sciences

CHARLES BEADNELL, Dictionary of Scientific Terms
Chambers Technical Dictionary
Condensed Chemical Dictionary
CRANE AND PATTERSON, Guide to the Literature of Chemistry
GRAY, Dictionary of Physics
JAMES, Mathematics Dictionary
KIRK AND OTHMER, Encyclopedia of Chemical Technology
McGraw-Hill Encyclopedia of Science and Technology
Science Abstracts
SINGER AND HOLMYARD, A History of Technology
SOULE, A Library Guide for the Chemist
Van Nostrand's Scientific Encyclopedia

Social Sciences

ADAMS, Dictionary of American History
American Labor Yearbook
Cambridge Ancient History
Cambridge Medieval History
Cambridge Modern History
DUTCHER, Guide to Historical Literature
Economic Almanac
International Encyclopedia of the Social Sciences

1.2.4 Catalog Card

Sample library card from the catalog.

301.36 S472b	Cities and towns -- Hist.	Subject card
Schneider, Wolf, 1925- Babylon is everywhere; the city as man's fate. Trans-		
301.36 S472b	Cities and towns -- Growth	
Schneider, Wolf, 1925- Babylon is everywhere; the city as man's fate. Trans-		
301.36 S472b	Babylon is everywhere	Title card
Schneider, Wolf, 1925- Babylon is everywhere; the city as man's fate. Trans-		
301.36 S472b	Sammert, Ingeborg, tr.	Translator Card
Schneider, Wolf, 1925- Babylon is everywhere; the city as man's fate. Translated from the German by Ingeborg Sammet and John Oldenburg. New York, McGraw-Hill, 1963, 400 p. illus. 22 cm. Includes bibliography.		
1. Cities and towns--Hist. 2. Cities and towns--Growth. I. Title.		
HT111.S313 301.36 63-17490 †		
Library of Congress (7-1)		

Call number	Author's name	Author's dates	
301.36 S472b	Schneider, Wolf, 1925-		Author Card
	Babylon is everywhere; the city as man's fate. Translated from the German by Ingeborg Sammet and John Oldenburg. New York, McGraw-Hill, 1963,		Translator, place of publication; publisher; date of publication
	400 p. illus. 22 cm. Includes bibliography.		
	1. Cities and towns--Hist. 2. Cities and towns--Growth. I. Title.		Other cards in catalog
	HT111.S313	301.36	63-17490 †
	Library of Congress	(7-1)	

Library of Congress call number Dewey Decimal System call number

Unit–2

TECHNICAL EXPOSITION

Written by: Mubadirah Durrani
Reviewed by: Syed Zafar Ilyas

OBJECTIVE

To develop various types of paragraphs with the help of topical arrangement, classification and definition.

2.1 WHAT IS A PARAGRAPH?

Now ask yourself

1. What is a paragraph?.....
.....
2. What is proper “punctuation” for a paragraph?
Where is it located in the paragraph?.....
3. How long should a paragraph be?.....

2.1.1 TOPIC VS TOPIC SENTENCE

The first sentence of a paragraph is usually called the *topic sentence*. You have no doubt noticed in your reading of certain English texts that it is possible to place the topic sentence at the end of the paragraph (as a kind of conclusion), or even the *middle* (as a kind of link between the two parts). It is even possible not to have a topic sentence at all; in the case, we say that the topic sentence is *implied or suggested*. You should be aware of these possibilities when you read; otherwise, you might miss the point which the author is trying to make. When you write, however, remember that a topic sentence placed at the beginning of a paragraph is the clearest kind of paragraph organization – simple, effective, easy for you to manage, and easy for your reader to understand.

What makes a good topic sentence?

The most important thing to remember at this point is the following: in a topic sentence, always try to make a statement about your topic which *limits* it to a certain extent:

Topic Sentence = TOPIC + LIMITING STATEMENT

Take, for example, the general topic of *soccer*. There are too many things to say about soccer to put into a single paragraph. Therefore, your problem as a writer

consists in deciding how you want to write about soccer. In other words, you need to *limit* your discussion.

One good way to limit your topic is to place *key words or phrases* in the topic sentence. These words or phrases will let the reader know how you are going to discuss the topic. These words or phrases are sometimes called “controlling words or phrases,” since they control the organization of the paragraph. In a paragraph on soccer, for example, they will immediately indicate to the reader that you plan to do *one* of several things.

Discuss the history of soccer
 Compare it with another sport
 Describe its difficulty
 Explain the rules of the game

How do you limit a topic in a topic sentence? There are many ways, but below you will find a list of the most common. Once you understand these examples, you will find it much easier to write a carefully controlled topic sentence?

Topic	Statement which Limits the Topic
1. Soccer	is now played <u>in the United States.</u> (geographical location)
2. Soccer	has become more popular <u>Within the last five years.</u> (time or period of time)
3. Soccer	is <u>a physically demanding sport.</u> (a certain aspect)
4. Soccer and football	<u>have a great deal in common.</u> (showing similarities)
5. Soccer	is <u>more dangerous than tennis.</u> (Showing differences)
6. A soccer player	can receive <u>various kinds of penalties</u> (a number of things; a list)
7. The World Cup Soccer Championship Games	<u>create interest</u> from soccer fans all over (effect) The world.
8. Soccer	is dangerous <u>for several reasons.</u> (cause; reason)

NOW ASK YOURSELF

1. Where does a topic sentence usually come in a paragraph?
 2. Where else can it come?
 3.
 4.
-
1. Zoos are popular with all children. They are able to see examples of wildlife from all continents. In wildlife preserves in Africa, the animals wander about freely without fear of being captured. Perhaps the favorite spot in a zoo is the elephant cage. There the elephants entertain the children by spraying themselves with their trunks and doing various tricks. The children are especially delighted when an elephant takes peanuts from them with his trunk.
 2. Editorials differ from other kinds of news stories. In most regular news stories, it is assumed that no personal opinions are being represented. Newspapers writers are supposed to present the facts of each story in a straightforward, unbiased fashion. Statistics reveal that over sixty-one million newspapers are sold every day in this country. The purpose of the editorial page, of course, is to allow the editors to give their personal opinions. Here is where they tell readers what they think about an issue – who is at fault, who has done a good job, or how a situation could be improved. It is no wonder, then, that Thomas Jefferson was once led to remark that he could more easily accept newspapers without government than government without newspapers.
 3. In order to find a suitable apartment, you must follow a very systematic approach. First, you must decide which neighborhood would be most convenient for you. Then you must determine how much rent your budget will allow. Utility bills for houses average between \$100 and \$150 per month. Your next step is to check the classified ads in the newspapers. If you do not plan to buy furniture, you should check under “Apartments for rent –unfurnished.” In either case, however, be sure to check these ads regularly, since new listings appear each day. After you have telephoned the apartments which seem likely choices, you must begin your long journey to inspect each one of them.

2.2 IDENTIFYING SUITABLE TOPIC SENTENCES

Directions: After reading each of the following paragraphs, select the most suitable topic sentence from the three choices following it, and write the sentence which you have selected in the blank. Then try to explain why each of the other two items is not appropriate. Follow the example. Possible reasons for not choosing an item might be that

It is too general;
It is too specific;
It is not a complete grammatical sentence;
It does not relate to the supporting sentences.

- I. There are some minor differences between American and British spelling. Where Britons end certain words with *-se*, Americans usually end the same words with *ce* (British *practise* vs. American *practice*); the reverse is sometimes true, too (British *defence* vs. American *defense*). Notice also the British preference for final *-re* over the American *-er* (*metre* vs. *meter*). Finally, most Americans consider *neighbor* a correct spelling, but a Briton characteristically adds a *u* and spells the word *neighbour*.

Choose a Topic Sentence

- A. British and American English are not the same.
B. There are some minor differences between American and British spelling.
C. The endings of British and American words are not the same.

Defend Your Choice

1. I did not choose letter A because it is too general (some aspects are not the same; other are).
2. I did not choose letter C because it is too general (some endings are not the same).

II.

.....

The operation of both can be thought of as being divided into three phases; *input processing*, and *output*. In the case of the computer, the information

which is fed into the machine – the data – is the input; the internal operations of the machine constitute the processing; and the result – usually a printout – is called the output. The telephone, too, acts on information presented to it and produces a result. The input is the actual dialing of the number. The switching system which locates the number can be considered the processing phase. Finally, the telephone rings on the other end of the line, indicating that the call has been completed; this constitutes the output.

Choose a Topic Sentence

- A. Both the computer and the telephone are helpful inventions.
- B. Computer terminology, such as *input* and *output*, is frequently used in other contexts.
- C. Despite apparent differences, the operation of the computer and the telephone have much in common.

Defend Your Choice

- 1. I did not choose letter _ because
.....
- 2. I did not choose letter _ because
.....
- III.
.....

The jack is a portable device for raising the car. It operates by means of force being applied to a level on which the car is balanced. The lug wrench is a tool with a fixed “jaw” for gripping the lug (the type of screw used to hold a tire in place). It has a long handle so that it is effective in turning the lug, either to tighten or to loosen it. These are the only two tools necessary to change a tire. They are usually found in the trunk of the car and are kept there at all times so that the motorist can use them, should he have a flat tire.

Choose a Topic Sentence

- A. Flat tires constitute a serious problem for the motorist.
- B. How to change a flat tire.
- C. Only two tools are considered standard equipment on new US automobiles.

Choose a Topic Sentence

- 1. I did not choose letter _ because
.....
- 2. I did not choose _ because
.....

2.3 SUPPLYING APPROPRIATE TOPIC SENTENCES

Directions: The topic sentences of each of the following paragraphs have been omitted. After a *careful* reading, write an appropriate sentence for each. Notice that the *general topics* of these paragraphs are the same as those in the previous exercise.

- 1.
.....

When you have removed the hubcap from the wheel which has the flat, the jack should be correctly placed so as to be able to lift the car off the ground. Now you are ready to jack up the car high enough for the tire to clear the ground. After you have done that, carefully loosen the nuts that hold the tire and rim in place; the tool which you use to do that is called a lug wrench. Remove the tire and put the spare tire in place. Now you are ready to put the nuts back on the wheel and tighten them well with the lug wrench. All that remains is to replace the hubcap.

- 2.
.....

The one most people are familiar with is the “desk dictionary,” sometimes referred to as general purpose dictionary. Another kind is the pronouncing dictionary, which is concerned with a word’s pronunciation more than with its meaning. A third type is the bilingual dictionary, which lists the words in one language and attempts to give equivalent meanings in another language. Other types include technical dictionaries, special purpose dictionaries, and scholarly dictionaries.

3.
.....

Under this new system, the customer’s monthly telephone bill includes specific information for each long-distance call: the date and time of each call, the rate charged per minute (based on the company’s discount system), the length of time the call took, the number and place called, whether the call was direct-dialed or operator assisted, and the amount charged for the call. As each call is placed, all of this information is fed into a computer and programmed onto each customer’s billing card, thus simplifying and clarifying the entire billing process. The Bell Telephone Company hopes that its new billing procedure will reduce the number of inquiries and free its employees to do work which computers are not yet able to perform.

4.
.....

To be eligible for these benefits, a person must have been participating in the Social Security system for a certain number of years. First, *retirement benefits* provide a worker with a monthly income ranging from \$230 to almost \$490, depending on his salary when he was working and the number of years he paid into the system. The second benefit is survivors’ benefits, a kind of life insurance which provides a deceased worker’s widow and children under eighteen with monthly cash payments based on what the worker would have received as retirement benefits. A third category is disability benefits. Workers, their widows, and their children under eighteen may be eligible for monthly payments if they are unable to work because of severe physical or mental illness. Finally, most people over sixty-five who are receiving disability payments are also entitled to Medicare. This insurance protection helps to pay hospital costs and covers

partial payment for other medical expenses, such as doctor bills and medication.

2.4 DEFINITION

In formal writing, it is sometimes necessary to write a paragraph to explain with a term means or how you are using it in a particular situation. This is called a paragraph of definition. A paragraph of definition may be either a formal definition, which explains the meaning as you might find it in the dictionary, or a *stipulated* definition, which explains how you are using a particular term within a specific context. In both cases, you will notice that definition often involves a combination of the kinds of development.

2.4.1 The Formal Definition

Model Definition

A wristwatch is a mechanical time-telling device which is worn on a hand about the wrists.

As you can see, a formal definition includes three kinds of things; the term to be defined, the class to which a thing belongs, and the features which distinguish it from other things in that class. In the case of a wristwatch:

Term = wristwatch

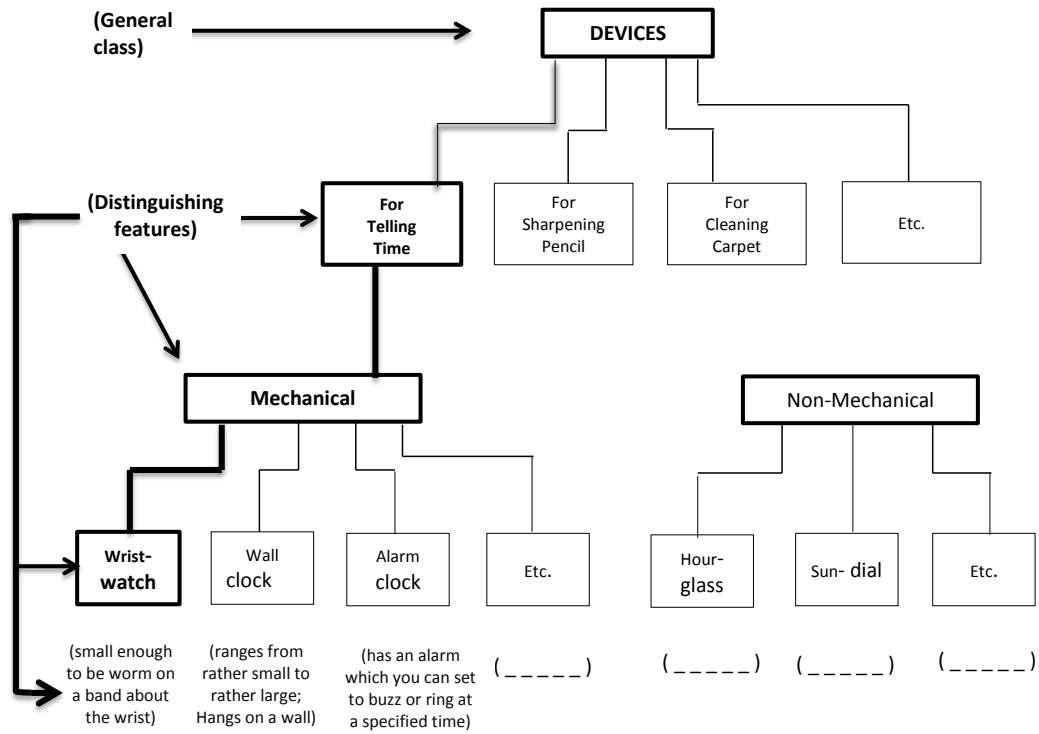
Class = device

Distinguishing features =

- 1) mechanical
- 2) for telling time
- 3) worn on a hand about the wrist

Thus, it is distinguished from non-mechanical time-telling devices (e.g., sundials) and other kinds of mechanical ones (e.g., alarm clocks).

A diagram of the information might look like this:



EXERCISE I

CATEGORIZATION OR CLASSIFICATION

Directions: Below is a chart with three terms. Fill in the chart for each of the terms. Then write one-sentence formal definitions.

TERM	GENERAL CLASS	DISTINGUISHING FEATURES
A cafeteria	Restaurant	Self-service
A pencil		
A rose		

2.4.2 SHORT PARAGRAPHS OF EXTENDED DEFINITION

When the general class and distinguishing features have been given, the writer may then go on to expand or *extend* his formal definition by giving additional information about the term being defined. This might include such things as physical description or a list of the advantages of the item. In the case of wristwatch, for instance, the writer might want to comment on variety in appearance and popularity. Notice the following paragraph on the wristwatch, where all of this information has been included.

A wristwatch is a mechanical device which is used for telling time. Its main advantage over other types of time-telling devices (such things as clocks, sundials, or hourglasses) is that it is small enough to be worn on the wrist, so that one can easily know the time by looking down. Wrist-watches come in various shapes and sizes, but all have one thing in common: a band or strap with which they may be attached to the wrist. In the United States, where “time is money,” practically everyone wears a wristwatch.

CASUAL ANALYSIS

Now Ask Yourself

In which sentence of the preceding paragraph do you find evidence of the following?

1. Enumeration?

2. Cause-effect?.....
3. Comparison-contrast?
4. Examples-details?

EXERCISE – 2

2.5 PARAGRAPH WRITING:

Definitional Information Transfer

Directions: Referring back to Exercise 1, choose two of the formal definitions which you wrote and develop each of them into a short paragraph of extended definition. Be prepared to explain how you have extended or expanded your original definition:

Purpose or use
Physical description
Various kinds or types
Advantages or disadvantages
Historically interesting information

2.5.1 DESCRIPTIVE WRITING

Different kinds of order may be used depending on what is being described. If you are describing a man, you will choose different things to say than if you are describing a mountain or a restaurant or a dress.

One kind of description follows SPACE ORDER. In space order you tell where things are. This is the order you will probably follow if you want to describe a place. Below is a description of a classroom.

Example:

The classroom is large, clean and well lighted. The walls are pale green. On the wall at the left as you enter there are three large windows. The teacher's desk is in

*the front. Blackboard covers
most of the wall at the right.*

This writer describes the room as it would look from the door – what is on the left what is in front, and what is on the right. The important thing is to have some order that will be easy for the reader to follow.

Below is another example of a description.

Example:

The man who opened the door in answer to my knock was an elderly man, white-haired and bent. He looked at me over his spectacles which were far down on his nose. In spite of his age, his dark eyes were keen and his voice was clear and strong. I noticed that he was wearing a bright-coloured sports shirt.

The writer has selected the man's most interesting and most outstanding features to describe. This description is quite different from a description of a place based on space order. The kind of description you use depends on what you are describing.

The need to describe people occurs fairly often, although most writers of language do not usually need to write long description to achieve their purpose. The informal description of other people, or even of oneself, is common. The following areas may all form part of a personal description.

Physical attributes: hair, eyes, complexion, facial shape, other facial features, build, height, weight, gesture, dress, and related descriptions: size, colour, style, decoration.

Ways of introducing stages in a process.

The	First	This process	Begins
	Second		Commences
	Subsequent		Finishes
	Following		Concludes
The	Last		
	Final		

Activity

Now describe the way you would cook your favourite dish, or describe the way a kite is made.

2.5.2 EXPOSITORY WRITING

Expository writing includes a research papers, report and personal essays of opinion. Expository skills are being used all the time; while giving directions, convincing friends, or parents into letting you do things, sharing ideas about movies, books etc. When writing an essay, you write about what you know, what you like, what you think and what ideas you have. Each person as his or her own opinion about things; how he feels or thinks about any particular subject. In expository writing opinions are important. In order to talk about your opinions you need to give the facts and impressions that led you to your opinion to make it real to your reader. In expository writing, you try to prove to your reader that your thoughts are worth while. You must therefore, give the reader your evidence. During our lifetimes we collect many facts, experiences and impressions.

Activity

Write a 450 word essay on the “Meaning of Democracy”.

You are expected to write a thesis statement, make an outline, write the first draft, discuss it with your colleagues, write the second draft, making all the necessary changes. This could be the final draft.

Vocabulary

The following words may be useful to you in talking and writing about the subject of democracy.

DEMOCRACY

Civil rights (always used in the plural) Individual rights, freedom of speech, the press, religion, assembly, etc.

Coalition: A temporary union of political parties to form a government usually found in a parliamentary democracy.

Diversity: A wide range of difference in customs, beliefs, attitudes, etc.

Municipal: Relating to the local government or institutions of the town or city.

2.5.3 NARRATIVE WRITING

The subjects of narrative writing can be:

1. Subjects that require you to write a story.
2. Subjects that require you to write an account of events.

In each kind of narrative the action provides, the chief interest, but a story has some special features that are not required in a straight forward narration of events.

An account of events: does not need a plot, but must progressively relate a sequence of events, a series of incidents. The writer must provide links to connect successive events and to keep the narrative flowing in an interesting way. The action must move steadily forward from the beginning to the end. A successful narrative depends, on a clear plan.

‘Characterisation’ (in the story sense) is not required, but some ‘human interest’ makes for an interesting narration. There can be some interesting glimpses of people taking part in the happenings but the emphasis, however, is on action, rather than character. Dialogue is not essential, but it can sometimes be

introduced to good effect, directly related to the action. Its purpose is to advance the narrative flow. An account of events should be rooted in the writer's own experience. First hand knowledge is needed. Details may be invented to add life and colour, but this kind of narrative composition is essentially an imaginative treatment of actual events. It is not piece of fiction, as a story description plays an important part in the narration. The emphasis is an action, but action does not take place in a vacuum. Details are needed to help the reader follow the events; interest is aroused by lovely descriptions of a realistic setting.

There are various ways to organize the sentences in a piece of writing. In narration, the sentences are usually organized according to time order. One thing happens and then another thing happens, and the events are told in the same order. Below are some words and phrases to help you give the impression of a time sequence.

A little while later...

During the ...

As the ...

Later than ...

Suddenly ...

Before ...

Activity

Write a narrative explaining what happened when you lost something.

2.5.4 DISCUSSIVE (ARGUMENTATIVE OR CONTROVERSIAL) WRITING

A discussive composition is one in which the writer presents facts, ideas and opinions about a given topic and arrives at a conclusion by reasoning. It is argumentative when writers are required to set out arguments on both order and to weigh them fairly. Some writing is controversial because it requires writers to keep cool and think clearly about topics that stir up strong feelings.

Before attempting an essay of this type, writers need to find facts and ideas on which to build an argument. There should be a genuine interest in the subject and an adequate fund of information about it; writers should have the ability to give due weight to a contrary opinion, while coolly and reasonably rejecting it. They should also be able to act out an argument step by step and to arrive at a sensible conclusion.

Writers need to plan a well-developed argument: in fact, the frame work of your essay should be reducible. Writers should use a clear, easily-read style so that attention is concentrated on what is being written rather than how it is written. The cool, thoughtful tone of good discussive writing calls for degree of formality in its style.

While writing such an essay avoid the use of emotional terms (e.g. hateful, shameless, ecstatic, disastrous etc) and excited writing.

The whole point of an argumentative essay is to win the reader over to your point of view. One way of being pervasive is to give examples to support your case. Don't generalize; the specific in your references.

Another way of giving vitality is to refer to what other people have said or felt about the matter. Quotations are particularly useful for beginning and ending an argumentative essay. They give life to what must other wise be pedestrian and flat.

Activity

Write an article on:

Hazards of Nuclear Power Station Near Habitation

Below is a list of slogans to help you think.

- Radio activity endangers our health.
- Radio activity poisons our earth.
- Nuclear wastes are a risk to life.
- We want no Chernobyl.
- Preserve natural beauty.

Try to turn each slogan into a statement.

Unit–3

ORGANISING INFORMATION AND GENERATION SOLUTION

Written by: Mubadirah Durrani
Reviewed by: Syed Zafar Ilyas

OBJECTIVE

To use outline and other brainstorming techniques to develop effective writing and utilize electronic devices for concise communication.

3.1 WRITING AN OUTLINE

Outlining is a useful skill to learn because it can be used in both your reading and your writing. In this chapter we will consider what an outline is and what it can be used for. We will then look at the form of two types of outlines. There will then be the chance to practice writing each type. In the following chapter on note-making, you will have additional practice in writing outlines of what you hear or read.

3.1.1 WHAT AN OUTLINE IS?

Basically, an outline is an organized list of related items or ideas. It is a method of grouping together things that are similar in some selected way, then presenting them in a simplified manner that clearly illustrates the relationship within each group and among all of the groups. For example, we can organize the following English first names into two groups: men's names and women's names.

Men's names

John
William
Richard

Women's names

Mary
Betty
Jane

Preparing an outline is essentially a problem of classifying and organizing. It is necessary to understand in what ways objects, facts, or ideas are related to each other.

3.1.2 WHAT AN OUTLINE IS FOR?

An outline is useful in both reading and writing. Because writing is language that is intended to be read, reading and writing are different ends of the same communication process. If a writer has used an outline to help him organize his writing clearly, his reader should be able to analyze its organization clearly by outlining it. In other words an outline is useful in planning the organization of writing. An outline is also useful in uncovering the organization of writing. The writer puts flesh on the skeleton of his outline. The reader removes the flesh to see the skeleton underneath.

For example, suppose your English teacher assigns a composition about traffic in the capital of your country. As you think about this topic, you decide that there are three main causes. You jot them down like this:

Traffic is a problem because
Too many cars
Narrow streets
Drivers don't obey regulations

Now you develop these ideas and write the following paragraph:

Traffic has recently become a serious problem in the capital of my country. There are three reasons. First, the number of automobiles has doubled in the past five years, with the result that there are more cars than the streets can accommodate. Second, the streets are old and narrow, because there are few garages or parking lots, these narrow streets are made narrower by the cars that are parked along the side. Third, many drivers do not obey the regulations, so that many traffic jams result from cars going the wrong way on a one-way street.

When the reader reads your paragraph, if he wishes he can make an outline of its organization. His outline might look something like this:

Traffic problem: three causes

1. Many cars
2. Narrow streets, made more narrow by parking
3. Many drivers do not obey regulations

From this example, you can see that an outline has two purposes:

1. In writing, to organize and present your ideas effectively
2. In reading, to analyze the organization and relationship of ideas.

3.1.3 HOW AN OUTLINE IS WRITTEN

An outline is usually written in one of two forms, depending on its purpose and its subject;

1. A topic outline
2. A sentence outline

Both have certain similarities which should be understood and used.

3.2 THE TOPIC OUTLINE

Here is an example of a topic outline. Compare the outline at the right with the simple list at the left.

The Plays of Shakespeare

Macbeth
The Comedy of Errors
Othello
Henry V
The Merchant of Venice
Romeo and Juliet
Love's Labor's Lost
Richard II
Henry IV, Part I
Hamlet

The Plays of Shakespeare

- i. *Tragedies*
 - A. *Macbeth*
 - B. *Othello*
 - C. *Romeo and Juliet*
 - D. *Hamlet*
- ii. *Comedies*
 - A. *The Comedy of Errors*
 - B. *Love's Labor's Lost*
 - C. *The Merchant of Venice*
- iii. *Histories*
 - A. *Richard II*
 - B. *Henry IV, Part I*
 - C. *Henry V*

This list at the left includes the same titles of plays as the list at the right. But, there is no organization in the list at the left. The plays are not listed in any meaningful way. At the right the titles have been outlined. They have been grouped according to similar characteristics under three main headings: tragedies, comedies and histories. This kind of organization is helpful in making the distinctions clear and in focusing on one particular distinction at a time.

3.2.1 FORMAL OUTLINE

Here is another example of a topic outline:

The Purpose of Study Skills

- I. To introduce students to proper attitudes toward their studies
 - A. Attitude toward English
 - 1. Reasons for studying English
 - 2. Study habits in English
 - B. Attitude toward other subjects
 - 1. The importance of motivation

- 2. The need for concentration
 - 3. The problem of distractions
- II. To instruct and to provide practice in study skills
 - A. Using a dictionary
 - B. Learning vocabulary
 - C. Outlining
 - D. Note-making
 - E. Using a library
 - F. Preparing for examinations

3.2.3 CONVENTIONAL OUTLINE

This outline is also a listing, divided into two main topics: I. to introduce students to proper attitudes, and II. To instruct and provide practice. This is also an example of conventional style of outline.

FORGETTING

- I. Why does forgetting occur?
 - A. Forgetting occurs because of disuse.
 - 1. Material that is not used is soon forgotten
 - 2. This is an old and no longer completely accepted theory.
 - B. Forgetting occurs because of interference.
 - 1. If you learn A, then learn B, when you try to recall A, B interferes.
 - 2. The learning of A interferes with the learning of B.
 - 3. Sleep provides a minimum amount of forgetting.
- II. How can forgetting be overcome?
 - A. Material should be meaningful and well organized.
 - 1. It is more difficult to memorize nonsense words than meaningful words.
 - 2. Material should be structured and organized in some meaningful way.
 - B. Material should be over learned.
 - 1. If you have learned A very well, and B only half as well, A will be recalled more easily than B.
 - 2. Do not be satisfied with only one complete, correct learning. Repeat the material over and over again.
 - C. Material should be systematically reviewed.

3.3 BRAINSTORMING

Brainstorming By Generating Questions

It is possible, of course, to expand such sets of questions. Here, for instance, is an expanded set of questions for 'Definition':

What is a Festival?

Is it an anniversary, a religious event, a kind of entertainment?

How many types of festival are there?

What is a festival related to?

How are the different types of festival related to each other?

It is good for students to create their own pool of questions which they find useful. Students can also keep a similar record in their workbooks under different headings, for example:

Things

What is A?

What is the colour
size, shape, etc., of?

Animals

What is A?

Is it an animal,
a bird, etc.?

How big is it?

Events

When did A happen?

What caused A?

and so on.

An even more extensive selection of questions can be used, covering a large range of functions, not all of which may be appropriate to the topic concerned. Make the questions available on a handout or in your notebook.

The questions are not, of course, in any kind of order and answers will need organizing and grouping in subsequent stages.

Questions

1. What does A mean?
2. How can A be described?
3. What are the component parts of A?
4. How is A made or done?
5. How should A be made or done?
6. What is the essential function of A?
7. What are the causes of A?
8. What are the consequences of A?
9. What are the types of A?
10. How does A compare with B?
11. What is the present status of A?
12. How can A be interpreted?
13. What are the facts about A?
14. How did A happen?
15. What kind of person is A?
16. What is my personal response to A?
17. What is my memory of A?
18. What is the value of A?
19. How can A be summarised?
20. What case can be made for or against A?

Function

Definition
Description
Analysis
Analysis of process
Recommendation
Analysis of function
Analysis of origins
Analysis of outcomes
Classification
Comparison
Evaluation
Interpretation
Reporting
Narration
Characterisation
Reflection
Reminiscence
Evaluation
Summary
Argument

3.3.1 ORGANIZING MATERIAL

An important way of organizing ideas, described by Michael Hoey in *On the surface of discourse* (1983), is as follows:

Situation

Problem

Response

Evaluation/Result

With variations, this organization forms the basis of much discursive and academic writing, and a set of questions derived from this sequence makes a useful prompt for writers:

Situation

What is the present situation?
How did it come about?
What are the characteristics?

Problem	Is there a problem? What is it?
Response	How can the problem be dealt with? What alternative solutions are there? What constraints are there on each possible solution?
Evaluation	Which of the solutions is likely to be the best? What would be the result of applying any of the solutions?

If the writer is reporting on past problems and solutions, the questions will be modified accordingly, for example:

What was the situation?
What were its characteristics?
Was there a problem?

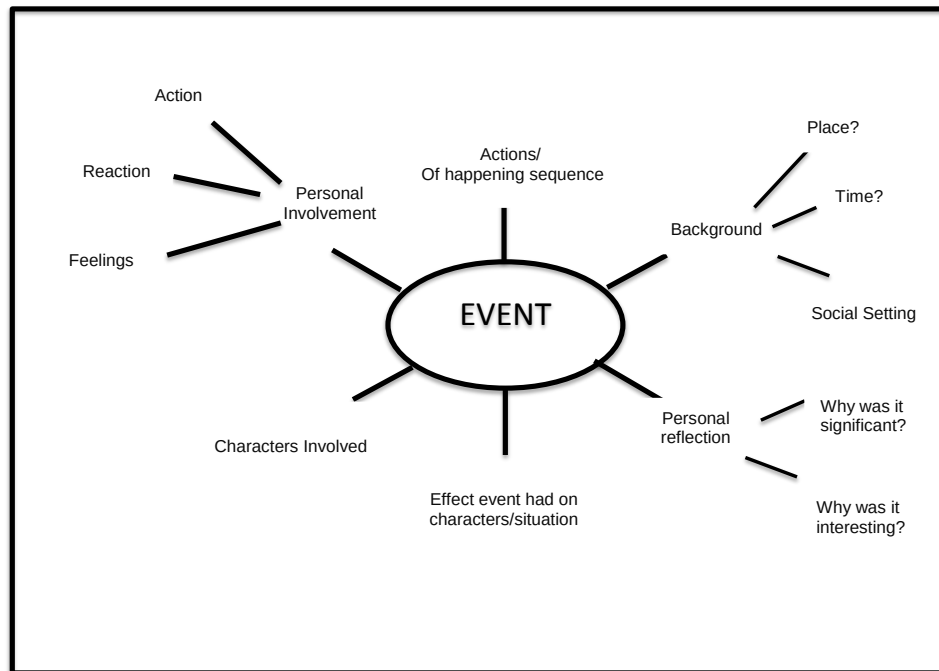
and so on.

Such an ordered set of questions defines the basic organization of ideas, thus providing a framework for structuring the actual writing. The questions can give rise to text ranging in length from a paragraph to a complete book.

Questions can also be used to stimulate the imagination by tapping the episodic and unconscious memory, particularly by drawing on the sense of hearing, sight, smell and touch. Using the sense helps to establish actuality, this is important in developing a sense of place and a credible story.

3.4 SPIDER-GRAM

Using the 'spider-gram' technique one can think, talk about the various elements which might be included in an account of a personal experience. It would probably look something like this:



3.5 ELECTRONIC COMMUNICATION

How to Email a Research Professor

Your email should:

- Have an informative subject line
- Be concise
- Be formal: Dear Dr. Smith; Sincerely, Your name
- Not use Mrs. Or Ms.
- Not have slang, abbreviations or emotions
- If applying for an opening:
 - address any qualifications the professor is looking for
 - demonstrate your experience
- If asking for a research opportunity;

- state specifically your interest in that research group (you need to read the professor’s website)
- explain why research is important for your goals
- ask to schedule a meeting or say that you will be coming to office hours

DO NOT SEND THIS EMAIL

Generalized from an email to a STEM Professor

Hi Joe,

My name is Name and I am a major in Major. Is there space in your lab for an undergraduate? If so, what is the pay rate?

Thanks,
Name

DO SEND AN EMAIL LIKE THIS

Generalized from an email to a UCSC Professor

Subject: Meeting to discuss undergraduate research opportunities in topic

Dear Dr. Professor,

I am a year student at university majoring in major. How you found out about the professor’s research. Expression of interest in specific paper or topic. I would appreciate the chance to talk with you about your research in topic of interest and about possible undergraduate opportunities in your lab.

My experience in research experience or class, confirmed my intention to develop my research skills and goal. I know you are busy. We could schedule an appointment or I can drop by your office hours on day and time.

I have attached my resume and unofficial transcript. Please let me know if there is any other information I can provide. I look forward to talking to you soon.

Best,

Name

Students often tell us they worry about how to address an e-mail message to a professor especially one whom they don't know. Below are suggestions that answer concerns we have heard not just from students, but from professors. And note; use these tips not just for e-mailing professors, but people who work in college offices, your employers and job supervisors, and your class deans and RDs.

3.5.1 On Addressing Your Professor:

E-mail to a professor should be treated like a business letter – at least until you know that professor's personal preferences very well. Although e-mail is widely regarded as an informal medium, it is in fact used for business purposes in many settings (including Wellesley College). You won't err if you are too formal, but there is the possibility of committing many gaffes if you are too informal.

The subject header should be informative. It is not a salutation line, so don't write something like "hey professor" in that line. Instead, write few words indicating the purpose of your message: "Request for a space in our class", for example.

Use professors' names when addressing them. Many professors we queried said that they do not like to be called simply "professor". They prefer "Professor Lee" or "Ms./Mr. Lee", most tell us that title itself does not matter nearly so much as the fact that you also use their names ("Dr. Lee" does seem to be uncommon at Wellesley, though, just so you know). Some professors will eventually suggest that you call them by their first names, but if you are more comfortable continuing to use a title, that is always fine. Just be sure to use a name. (Note: these comments are true for personal interaction as well as for e-mail).

Dear, Hi, Hey, or nothing? To some eyes and ears, "Dear Professor Jones" may be too formal for an e-mail message – but in fact it will do just fine when your purpose is a business-like one. Simply writing "professor Jones" (followed by a comma) is fine, too. Some faculty are sensitive to the word "Hi" as a salutation, whether alone or with a name (e.g., "Hi, Professor Jones"), but others don't mind it and in fact use it themselves. But avoid "hey" – no one we queried likes that one.

Don't expect an instant response. Although we have all become accustomed to the instantaneous quality of electronic communication, your professors want you to know that they simply cannot always answer a message quickly. Allow them a day or two, or even more, to respond. You can re-send the message if you have not heard back in five days or so.

One E-mail Style

Do not use smiley faces or other emoticons when e-mailing professors, and do not use all those internet acronyms, abbreviations, and shortened spellings (e.g. LOL, or “U” for “you”). Similarly, do not confuse email style with txt style. All of that electronic shorthand signals a level of intimacy (and perhaps of age) that is inappropriate for exchanges with your professors.

Write grammatically, spell correctly and avoid silly mistakes, Proofread, use the spelling checker. Especially double-check for embarrassing errors in your subject header. Show that you are about how you present yourself in writing in your professor.

Use paragraph breaks to help organize your message. It’s hard to read a long unbroken stream of words on a screen.

Keep most messages to under a screen in length; lots of readers will simply defer reading long message, and then may never come back to them. On the other hand, a very short, tense message may simply be meaningless. Be sure to include enough information so that your reader can understand what you are requesting. Provide a bit of background or context if necessary. State your request clearly.

Take extra steps to minimize the e-mail exchange; for example, if you are requesting an appointment, state your purpose and name the times that you could come in your initial message. Your respondent may then be able to answer you with only one additional message.

Quote selectively and briefly from any prior message to provide context and background. Although sometimes it’s good to quote an entire exchange so as to keep a record of what’s been said and decided, often that’s unnecessary and simply ends up making a message too long and cluttering the screen.

Many professors advise that you think about why you are sending an e-mail message. Are you asking something that could easily be checked if you took a few extra steps yourself? For example, e-mailing a professor simply to ask when her office hours are can be annoying when the office hours have been clearly announced on the syllabus already. On the other hand, e-mailing for an appointment is just fine. Are you asking a question privately that might be better asked on the course conference, where all the students might usefully see the response? Are you e-mailing to lodge a complaint or to ask for a letter of recommendation or to seek help with a problem set? In these cases, personal

contact and an office visit might be much better.

Be respectful, and think about what kinds of things might sound odd or offensive to your professor. For example, do not say flippantly that you slept through that professor's class, or talk about your love life, or bash chemistry or math or writing.

3.5.2 A FEW CONVENTIONS AND COMMON CONFUSIONS IN SCIENCE WRITING

Names of organisms. Biological organisms have an official scientific name (called a Latin Binomial), as well as one (to several) common names. Common names vary regionally, so clear communication requires the use of scientific names. Scientific names come in two parts (thus binomial), the first part the genus (plural, genera), the second part the species. The genus name is always capitalized and species name is never capitalized (even if it is a proper noun!) The genus and species names are always italicized or underlined. Recent changes in the nomenclatural code also require family name to be italicized, but this practice is not universally adopted (they should always be capitalized, however). Following the species name, one sometimes finds additional words. They may be a designation for a subspecies or variety (designated by a non-italicized "sap" or "var" followed by the italicized name), or may designate the "authority", the person who gave the organism its name. The authority is never italicized, and should be given with the organism's name the first time it is used in a paper, but not thereafter. The first time that the organism is named in a paper the full genus and species must be written out; after that, the first letter of the genus name, followed by the full species name may be used. Common names should be capitalized, but never italicized.

Coast Redwood
Sequoia sempervirens (D. Don) Endl.

common name
Full genus and species
binomial, followed by the
authority

S. sempervirens

abbreviated binomial to be
used after first reference

An Example:

Sequoia sempervirens (D. Don) Endl. (Coast Redwood, in the family *Taxodiaceae*) is the most outstanding tree on the UCSC campus. Restricted to the

fog zone along the Pacific coast, *S sempervirens* can grow to heights of more than 100m. Often growing along streams in redwood forests is the endemic California Hazel, *Corylus cornuta ssp. Californica* (A. DC.) W. Sharp.

3.5.3 Common Abbreviations

Et al. abbreviation for Latin *et alia* – others. Used when referring to studies done by three or more authors.

“In a landmark paper, Press *et al.* (1994) showed...”

i.e. abbreviation for Latin *id est* – that is. “Disturbed by irregularities in the 2000 presidential elections (*i.e.*, *the* Florida vote count), voters turned to...”

e.g. abbreviation for Latin *exempli gratia* – for example. “The state of California spends million of dollars each year to control its most noxious invasive weeds (*e.g.*, Scotch Broom, Salt-Cedar)...”

sp. Singular abbreviation for species. “Various weeds, including *Bromus sp.* Were...”

spp. Plural abbreviation for species. “Several *Brassica spp.* Are invasive weeds on the UCSC campus.”

3.5.4 Commonly confused words

“affect” is a verb. – “The heavy rains could affect the timing of migration this year.”

“effect” is a noun – “The effect of the heavy rains on migration could be dramatic.”

“Data” is plural (Singular, Datum). Data ARE, Data WERE; never Data is or Data was.

3.5.5 Parts of a Scientific Reference

Exact formatting of bibliographic references varies among journals and publishers; each journal will provide an “instructions to authors” in the first issue of each year (as well as on their website) detailing the structure of a reference. A few general rules are usually followed, however.

In the text, if there are one or two authors, used the names, followed by the date.

“Delving deeply into the ageless question ‘why whine about wine?’ Press (2001) uncovers long-obscured truths about the nature of yeasts.”

“Plant diseases are an essential component of tropical forest dynamics and conservation planning (Gilbert and Hubbell 1995).”

If there are more authors, use *et al.* (in italics) after the first author. “Rabkin *et al.* (1999) brought together disparate views on the natural history of nature writers into an insightful synthesis.”

In the bibliography, references should be in alphabetical order by author, and within author in order of date.

Holl, K. D. and R. B. Howarth. 2000. Paying for restoration. *Restoration Ecology* 8:260-267.

Holl, K. d. and M. Kappelle. 1999. Tropical forest recovery and restoration. *Trends in Ecology and Evolution* 14:378-279.

Interpretation of the citations: In the Holl and Howarth (2000) citation, the article titled “paying for restoration” was published in the journal “*Restoration Ecology*” in the year 2000. The volume of the journal was “8”, page numbers 260-267. Most journals also have an issue number within a volume (volume numbers change yearly, issue number designates the month or week within the year). This is sometimes written as *Restoration Ecology* 8(2)260-267 if the article were in the second issue; most journals, however, do not include the issue number.

Remember that every reference in the text must be included in the bibliography, and every reference in the bibliography must be used in the text.

3.5.6 Table and Figures:

In the natural sciences, the convention is always to place figures titles below the figure; place titles for tables above the table. Conventions in social sciences and the humanities vary among publisher. Follow the conventions of your audience. In all cases, the title should be descriptive of the table or figure, provide enough information that it can stand alone, separate from the text. Number all tables and figures, and refer to them in the text.

Unit–4

POLISIING STYLE

Written by: Mubadirah Durrani
Reviewed by: Syed Zafar Ilyas

OBJECTIVE

To utilize enumerator, listing signals and suitable vocabulary to polish writing skill.

4.1 ENUMERATORS

Notice the use of the word kinds in the preceding model paragraph. We will call this word an *enumerator* since it helps us show the reader exactly what we are listing or enumerating. In the model paragraph, it is kinds of materials. Remember that enumerators are valuable key words. You should try to put them in topic sentences of enumerative paragraphs. This will help you to organize your paragraph more clearly; it will also help the reader to follow your train of thought more easily.

Writers frequently wish to make a list of other things besides kinds or types. They may, for example, want to talk about

Classes
Parts
Elements
Factors
Characteristics
Aspects
Divisions
Subdivisions
Categories

4.2 LISTING SIGNALS

When making a list, people often use numerals (e.g. 1, 2, 3, etc.) to indicate the various items in the list. A simple list of this kind could be made for the model paragraph on library materials:

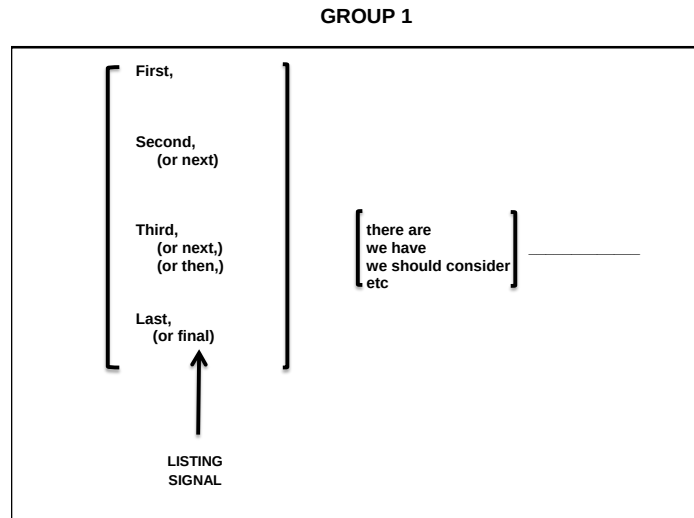
Kinds of library materials found in most good libraries:

1. Books
2. Reference works
3. Periodicals

In most formal writing, however, a list is usually not made with numerals. The items are indicated by what we call listing signals. The author of the model

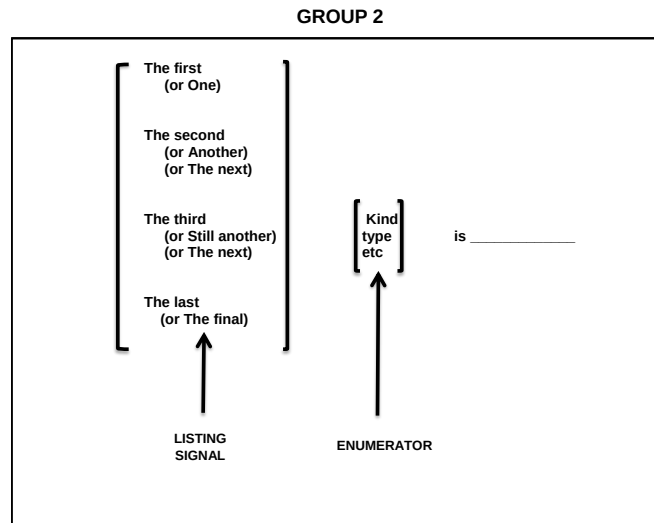
paragraph has used three of these listing signals: *first, ...; second, ...; third, ...*
 There are two main groups of listing signals in English. You should become very familiar with the words in these two groups:

GROUP 1



* A major exception to this is *scientific* and *technical* English, where it is common to find lists with a numeral before each item on the list.

GROUP 2



NOW ASK YOURSELF

1. What is the difference between an enumerator and a listing signal?
.....
.....
2. What is another way to say each of the following?
(Last,)
(The final kind)
(Another kind)
(Next,)
4. What element does a Group 2 sentence have that a Group 1 sentence does not?
.....

GROUP 1 VS. GROUP 2: SENTENCE STRUCTURE

There is one thing that you must remember. The words in Group 1 take a different sentence structure from the words in Group 2. Let's take a sentence from the model paragraph on library materials and illustrate this difference, which is really a very simple one:

Group 1:

First	there are	books.
-------	-----------	--------

Group 2:

The first kind	[is consists of]	books.
----------------	------------------	--------

Note that a full sentence follows the listing signals from Group 1. Notice also that in Group 2, the listing signal and enumerator are the subject of the sentence. These differences, although simple, are very important!

EXERCISE

Paraphrasing Listing Signals

Directions: Rewrite each of the following sentence in three (3) ways, choosing words from both groups of listing signals. Circle the enumerator if you have used one.

1. Still another kind is periodicals.
 - a. The next kind is periodicals.
.....
 - b.
 - c.
2. Last, there are reference works.
 - a.
 - b.
 - c.

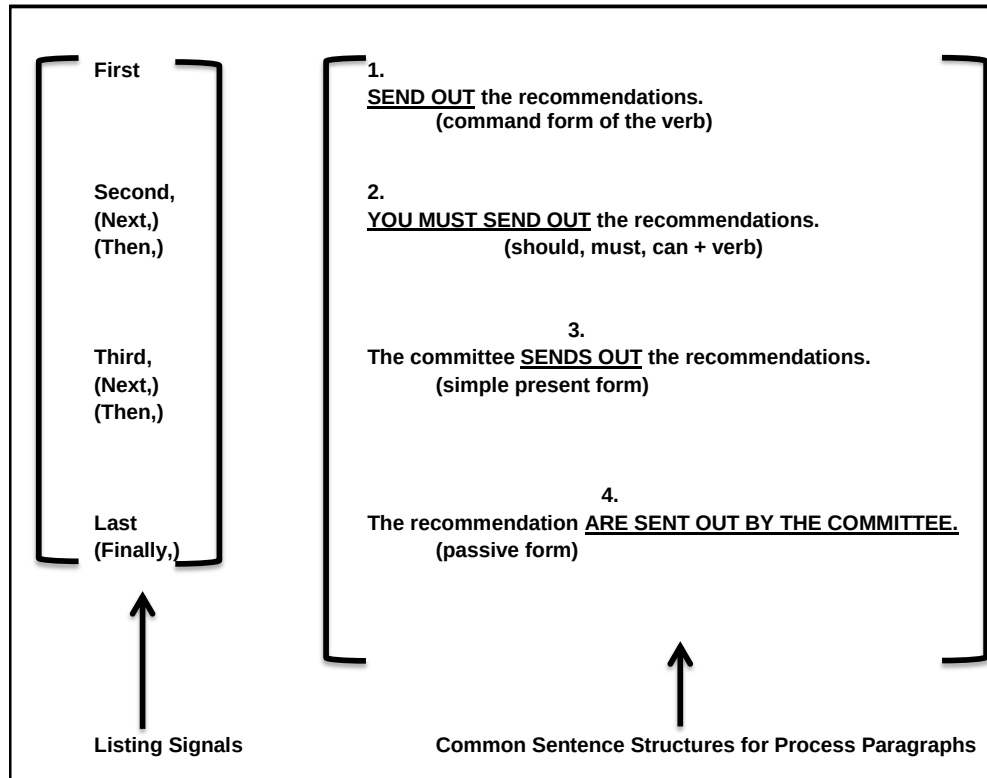
NOW ASK YOURSELF

1. In the paragraph on finding a suitable apartment (p. 16), do you notice any of these enumerators?
2. In the paragraph on changing a flat tire (p. 19), do you notice any of these enumerators?

LISTING SIGNALS

Process paragraph often use listing signals, Group 1 listing signals can be used, but the structures which follow them are usually different:

Group 1 (PROCESS)



NOW ASK YOUR SELF

Which of the above structures do the following sentences represent – command form, modal auxiliaries, simple present, or passive?

(from “Changing a Flat Tire,” p. 19)

1. The jack should be correctly placed
2. ...carefully loosen the nuts... ..
3. Remove the tire and put the spare tire in place.

(From “finding a Suitable Apartment.” P. 19)

4. ... you must follow a very systematic approach
5. First, you must decide which neighborhood... ..
6. ... you should check under... ..
7. Otherwise, check under... ..

Group 2 listing signals on (p.49) can also be used to develop process paragraphs:

Activity

Write an essay of 400-500 words on the topic of “Capital Punishment”.

4.3 VOCABULARY

The following words may be useful to you in talking and writing about the subject of capital punishment.

Capital punishment or the death penalty by law, punishment by death for a serious crime. The execution (putting to death) in the United States may be by the electric chair, firing squad, hanging or lethal (causing death) injection.

Kinds of Crime

homicide Killing, murder.

Arson: The intentional burning of property (arsonist, noun).

Rape: Sexual attack on person (usually a woman) (rapist noun).

Hijacking: Stealing goods from a moving vehicle (hijacker, noun)

Skyjacking: Taking over an airliner (skyjacker, noun).

Terrorism: The use of violence to achieve political goals (terrorist, noun)

Espionage: Spying (spy, noun).

Drug trafficking: Dealing in drugs (drug trafficker, noun).

Kidnapping: Seizing a person by force and holding that person for ransom or some other demand (kidnapper, noun).

Burglary: Breaking into a building, usually to steal (burglar, noun).

Robbery: Stealing, with the use of force (robber, noun)

Assault: A violent attack.

Treason: Betrayal of one's country by giving away its secrets (traitor, noun).

Sabotage: The deliberate damaging of property or the obstruction of production in a factory (saboteur, noun).

Degree of Crime

Offense: A violation of law or rule, a crime (offender. Noun).

Felony: A serious crime, punishable by a heavy sentence for example. Murder, arson (felon, noun; felonious, adjective).

Misdemeanour: A less serious offence than a felony.

Being Charged with a Crime

Arrest: Seize a person by legal authority.

Bail: Security (money or property) given as a guarantee that an accused person will return for his or her trial.

Grand jury: A group of citizens that determines whether there is enough evidence to bring an accused person to trial.

Indictment: Being charged by the law with an offense.

Arraignment: The act of summoning an accused person before a court because of the charges in an indictment.

Prosecute: To institute and conduct a legal action against an accused person. The prosecuting attorney (for prosecutor) as a representative of the law, makes the charges against the accused person in a trial.

Trial: A formal hearing to decide a case in a court of law. For a serious case, a trial includes the judge, the jury (in the United States, usually twelve citizens, who will decide whether the accused person is innocent or guilty), the accused person, the defense attorney, the prosecuting attorney, and witnesses for and against the prosecution.

Verdict: The decision of the jury regarding the innocence or guilt of an accused person in a court trial. If the verdict is guilty, the judge usually gives the sentence (the punishment) for the accused.

Convict: (verb) Find a person guilty of an offense (a convict is a prisoner)

Parole: the conditional release of a prisoner before his or her sentence ends. A parolee must report to parole officer.

Deterrent: Something that discourages a person from doing something for fear of punishment (deterrent, adj, deterrence, noun, deter, verb).

Rehabilitation: Restoration to useful life (especially to acceptable social behavior) through education and therapy.

Mandatory: Required by law (without exception).

Discretionary: Having the power to make decisions according to one's own judgment (thus allowing for exceptions).

Mitigating circumstances: Circumstances that contribute to making a punishment less severe.

Retribution: A punishment that is deserved (Because the word can be interpreted as a just punishment administered by a higher authority. It has a good or neutral connotation).

Revenge: Punishment inflicted in return for some harm, often personal harm (Usually, in discussions of capital punishment. Revenge has a negative connotation, as resulting from our lower instincts.)

Capricious: Tending to change one's mind unpredictably for no apparent reason.

Arbitrary: Determined by individual judgment rather than by a guiding principle or law.

Opponent: A person who is against something or someone.

Proponent: A person who is for something.

Inhuman: lacking pity, brutal, heartless.

Inhumane: Not caring for others, without sympathy.

Unit–5

DOCUMENT DESIGN

Written by: Mubadirah Durrani
Reviewed by: Syed Zafar Ilyas

OBJECTIVE

To bring awareness about essential format of writings to develop reports, manuals and thesis.

5.1 DOCUMENT STRUCTURE

Most of your reading will be the purposeful, study-type reading of textbooks. This includes facts, formulas, dates, causes, effects, attitudes, ideas, experiments, results – the whole range of information that is contained in the textbooks of your various courses. Though the subject matter is different, its organization and presentation are similar. It will be useful to look closely at what a textbook contains. It has a format or structure that can help you to design other forms of writings like reports, manuals and thesis. Let's go through the parts of a book.

5.1.1 TITLE PAGE

It is surprising how many students do not know the title of their textbook or its author's name. It may not seem important, but since you are going to put confidence in the author and his book, it may be worthwhile to know his name, who he is, and where he comes from. This is essential of course, if you must prepare footnotes or a bibliography.

5.1.2 PREFACE OR FOREWORD

In the preface (sometimes called the foreword or introduction), the author explains the purpose, organization, method of presentation, and whatever particular features of the book he wishes to call to your attention. Frequently in textbooks there is section titled "To the Student". Read it carefully. The author is recommending how to use his book efficiently.

5.1.3 TABLE OF CONTENTS

Skim through the table of contents to get an overall view of the material in the book. Some tables of contents are actually outlines, with subtopics of each chapter. This section of the text is the fastest, easiest way to survey the territory over which you will travel.

5.1.4 TEXT

The chapters of most textbooks include a variety of study guides. Section headings, in boldface type, announce the general subject of the material that

follows. Frequently you will find summaries at the end of each chapter, often with thought questions and exercises. Many texts contain maps, charts, diagrams and tables. Don't ignore them. They have been included to help you visualize the information. They are valuable aids to understanding. Know how to read them, and read them.

5.1.5 GLOSSARY

Many textbooks include a glossary, either at the end of each chapter or at the end of the book. This is a kind of dictionary which defines or explains some of the technical terms in the book, and often provides examples and page references.

5.1.6 BIBLIOGRAPHY

An alphabetical list of relevant books and articles is frequently included either at the end of each chapter or at the end of the book. If you want further information on the subject, or if you want to know the author's sources, the bibliography will list them by author, title, publisher and date.

5.1.7 INDEX

One of the most important sections of any textbook is the index at the end. This usually a thorough, fairly detailed alphabetical listing of all the major persons, places, ideas, facts or topics that the book contains, with page reference. For review, or for quickly locating some point you wish to check, the index is a valuable guide.

5.1.8 BIBLIOGRAPHICAL INFORMATION

For each book, write the name of the author(s), title of the book, place and date of publication, and the publisher.

5.2 FRONT MATTER

Does the book have a preface, foreword, introduction, To the Teacher, or To the Student? What information here might be useful for a better understanding of the purpose, content and arrangement of the text? What differences in this part do the two books show? Is one preferable to the other? Why?

5.2.1 TABLE OF CONTENTS

Does the Table of Contents give you a general idea of the material and organization of the book? Is it only a list of chapter titles? Is it a kind of outline with topics and subtopics? How do the two Tables of Content compare? Which is more useful? Why?

5.2.2 TEXT

Are there different typographical devices to mark off sections and to emphasize important material? Are there summaries, notes, or additional readings at the end of the chapters? Are there maps, diagrams, charts, graphs, or tables to clarify significant points? Is the type clear, the margin wide, and the overall effect of the page pleasing to the eye? Without considering their content, which of the books is more appealing because of its organization, layout and various reading aids?

5.2.3 BACK MATTER

Does either of the books have a glossary? Are technical terms defined clearly as they occur in the text? Are they defined in a separate section, either in the chapter or at the end of the book? Does either of the books have a bibliography? If so, does it have brief comments about the content and relative value of the books that are listed? Is the bibliography at the end of each chapter, or is it placed at the end of the text?

5.2.4 INDEX

Compare the index of each book. Is one more complete than the other? Is there special information at the beginning of the index that you should know about in order to make best use of it? Think of a specific topic in each course and try to find page reference quickly. Is there any reason to prefer the index of one book to that of the other? Why?

Below is the Table of Contents from a book by Robert A. Hall, Jr., entitled *Linguistics and Your Language*. Look at the closely, then answer the following questions.

CONTENTS

PART I:	THINGS WE WORRY ABOUT	
1.	Which Should I Say?	1
2.	Right vs. Wrong	9
3.	Marks You Make with Your Fist	30
4.	What Price History?	49
PART II:	HOW LANGUAGE IS BUILT	
5.	Language Has System	57
6.	Language Has Sound	68
7.	Language Has Form	97
PART III:	LANGUAGE IN THE WORLD AROUND US	
8.	Language Has Meaning	121
9.	Language Covers Territory	135
10.	Language Does not Stay the Same	157
PART IV:	WHAT WE CAN DO ABOUT LANGUAGE	
11.	Learning Your Own Language	191
12.	Learning Another Language	209
13.	One World, One Language?	228
14.	There's Nothing Wrong with Your Language	243
	Appendix A: Some Useful Books	260
	Appendix B: Additional Phonetic Symbols	264

1. This book is divided into four parts. How many chapters does the book contains?

.....
.....

2. Judging from the title of Parts I, II and III what is the general topic we worry about, referred to in Part I?

.....
.....

3. Assume that you are interested in each of the topic listed at the left below. If this book includes information about the topic, write the chapter number on the line at the right.

.....

.....

INDEX (Sample)

Entries relate to Sections One, Two and Three of the text, and to the glossary. References to the glossary are indicated by 'g' after the page number. Titles of publications are given in *italics*.

- academic learning skills 51–2, 102–4
- 'accumulated entities' view of language learning 29
- acquisition *see* SLA (second language acquisition)
- activities, classroom 149–57
 - Bangalore Project 43–4
 - focus of process-oriented objectives 131–2
 - sequencing 112–15
 - typology 55–7
- adaptation by teacher 139
- ALL (Australian Language Levels) Project 81, 87
- analytic syllabus(es) 37–9, 158g
 - planning 27–8
- Anderson, A. 60, 133
- Australia, content-oriented ESL syllabuses 49
- Australian Adult Migrant Education Program 80

- Bailey, K. 32
- Bangalore Project 42–4, 99–102
- basic personal communication skills 51–2, 102–4
- biographical information, evaluation of use 140
- Breen, M. 6, 43, 52, 54
- 'British' school of applied linguistics 40
- Brown, G. 58–9
- Brumfit, C. J. 36, 40, 87, 99–100
- Burt, M. 32

- Cambridge English Course, The* 81–3, 86, 89–90, 92
- Candlin, C. 3, 45–6, 59

- choice, in knowledge framework 49, 50, 105
- classification, in knowledge framework 49, 50
- code complexity 59
- cognitive load 59
- communication skills
 - for different areas of use 21
 - natural approach 51–2, 102–4
- communicative approaches 158g
- communicative event, in needs analysis model 20, 75, 76
- communicative function(s) *see* function(s)
- communicative goals 25, 81, 96–7
- communicative key, in needs analysis model 20, 75, 76
- communicative stress 59
- communicative syllabuses 11
- conditions, relation to tasks and standards 125, 155
- content
 - experiential 38, 147–9
 - see also* knowledge framework
 - grading 85–95, 148–9
 - reformulation as objectives 154–5
 - vs. methodology 5–7
- content-based syllabuses 38, 48–51, 104–7
- content continuity 59
- content selection 10–26, 68–9, 85–95, 147–8
- content survey form 147–8
- coursebooks
 - grammatical items 146
 - relation to objectives 127–30
 - task grading 152–3
- 'cracking the code', grammatical syllabuses 29

GLOSSARY (Sample)

analytic syllabus: a syllabus based on non-linguistic units such as topics, themes, settings, and situations. Learners are exposed to holistic 'chunks' of language and are required to extract patterns and regularities from these.

communicative approaches: approaches to language teaching in which the focus is on processes of communication rather than on structural, functional, or notional items.

curriculum: principles and procedures for the planning, implementation, evaluation, and management of an educational programme. Curriculum study embraces syllabus design (the selection and grading of content) and methodology (the selection of learning tasks and activities).

function: the communicative use to which an utterance or longer piece of language is put. Examples of functions include: apologizing, greeting, describing, defining, contradicting.

goal: the broad, general purposes behind a course of study. Goals can be couched in terms of what the teacher is to do or what the learner is to do. Examples of goals:

'To develop conversational skills.'

'To develop skills in learning-how-to-learn.'

'To teach learners basic grammatical structure.'

'To prepare learners for tertiary study in a foreign language.'

grading: the arrangement of syllabus content from easy to difficult.

methodology: the study and development of learning tasks and activities.

needs analysis: techniques and procedures for obtaining information from and about learners to be used in curriculum development.

notion: the concepts expressed through language. Examples of notions include: time; frequency; duration; causality.

objective: a statement describing what learners will be able to do as a result of instruction. Formal objectives are meant to have three parts: an activity (what learners will do); conditions (under what circumstances), and standards (how well they will perform).

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Unit–6

CITATION

Written by: Mubadirah Durrani
Reviewed by: Syed Zafar Ilyas

OBJECTIVE

To enhance knowledge about report structures and citation styles to produce quality work.

6.1 CITATION STYLES

There are many styles for citing references. Although strict standards (e.g. British Standards) for citing references exist, my advice is not to bother with them; instead, find a reputable journal in the library and copy in style. Alternatively, copy the example below. It is important to be consistent, complete and unambiguous; beyond that, it doesn't matter much what you do.

Example citation style:

Citation in text:

Mander, in "Notes on a system specification method" [Mander 1983], gives the following...

... as described by Briggs [1983a]...

Thimbleby's guidelines [Thimbleby 1983] suggest that...

Different methodologies have been examined [Tully 1983].

Several recent publications in this field [Wand 1980d, ACM 1971] have been very influential.

At the end of the report list of references is provided that you can see on next page.

6.2 REFERENCES

- ACM 1971. Association for Computing Machinery, *Second symposium on problems in the optimisation of data communication systems*, ACM (1971)
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If you adopt this style, when you cite a reference, you need not repeat the author’s name or authors’ names (“Jones and Sanderson [Jones & Sanderson 1999] have shown...”). Write instead: “Jones and Sanderson [1999] have shown...”, and list the reference as “Jones & Sanderson 1999”.

6.3 REPORT STRUCTURE

Top-Level Structure

At the top level, a typical report is organised in the following way.

1. Abstract. (This is a couple of paragraphs – no more – which summarizes the content of the report. It must be comprehensible to someone who has not read the rest of the report.)
2. Introduction. (The scope of the project, setting the scene of the remainder of the report.)
3. Previous work. (One or more review chapters, describing the research you did at the beginning of the project period.)
4. Several chapters describing what you have done, focusing on the novel aspects of your own work.
5. Further work. (A chapter describing possible ways in which your work could be continued or developed. Be imaginative but realistic.)
6. Conclusions. (This is similar to the abstract. The difference is that you should assume here that the reader of the conclusions has read the rest of the report.)
7. References and appendices.

References

References must be relevant. A typical PR3 project report might contain about one page of pertinent references, if the initial research period was well spent. Do not include references.

Alternatively, a system of numbered references, such as the default format produced by the Unix refer tool in conjunction with troff, is acceptable. I myself much prefer numbered citation styles, which I find much less obtrusive and easier on the eye; e.g. “Jones and Sanderson have shown ...” or “Jones and Sanderson [1] have shown...”. These forms, which are allowed by the regulations in the Handbook, seem to be the two dominant citation styles in academic journals.

You may wish to refer to electronic sources, particularly material found on the World-Wide Web. It is not enough to put “found on WWW” in place of citation. The web page [“Bibliographic Formats for Citing Electronic Information”](#) gives advice on citing on-line sources.

If possible, avoid citing unpublished literature. It is however, acceptable to cite university reports, such as this Department’s YCS series, and PhD theses (although getting hold of the latter can be almost impossible).

“References” are always cited in the text. Other works you have made use of but not cited should be listed in a section called “Bibliography”.

Note that “et. al.” requires a period after the abbreviation “al.” (for “alia”). It means “and others”, and may be used only to refer to people, typically in lists of references. It is the animate form of “etc”, which also requires a period.

6.4 LOWER-LEVEL STRUCTURE

Structure is a recursive concept. A well-structured report has its top-level section well ordered, and it is easy to get this right; but each section must in itself be well ordered, and that is more difficult.

Most paper documents, and many on-line documents, are read linearly from beginning to end. This is certainly true of an examiner reading a project report. Consequently, the writer of a well structured document avoids forward references wherever possible. Try to avoid writing “...as we shall see in chapter 10,...”, especially if the material in chapter 10 is essential to an understanding of the text at the point where the reference occurs. Occasionally such reference are unavoidable, but more often than not they are a sign that the text needs to be re-ordered.

In the old days, re-ordering text entailed “cutting and pasting” with real scissors and real paste. Nowadays, the word-processor has made these operation so easy that there is not excuse for slovenly structure. Take your time, and keep rearranging words or phrases within sentences, sentences within paragraphs, paragraphs within sections and section within the whole report until you have got it right. Aim for a logical progression from beginning to end, with each sentence building on the previous ones.

If the chapters are numbered 1, 2, 3, ..., then the sections within (say) chapter 1 will be numbered 1.1, 1.2, It is permissible to sub-divide a section: the sub-section within section 1.1 will be numbered 1.1.1, 1.1.2, Do not however, nest sub-sections to more than four levels: sub-sub-section 1.2.3.4... is acceptable, but 1.2.3.4.5 is not. It is quite possible, with care, to write even large and complex books without using more than three levels.

6.5 FOOTNOTES

Footnotes are a nuisance to the reader. They interrupt the linear flow of text and necessitate a mental stack-pushing and stack-popping which demand conscious effort. There are rare occasions when footnotes are acceptable, but they are so rare that it is best to avoid them altogether. To remove a footnote, first try putting it, in-line, surrounded by parentheses. It is likely that the poor structure which was disguised by the footnote apparatus will then become apparent, and can be improved by cutting and pasting.

YOU AND YOUR SUPERVISOR

Writing is a solitary pursuit. Whereas your supervisor will guide you through the early stages of your project work, you must write the report on your own. It is University assessment, and the rules on plagiarism and collusion (do consult the Student's Handbook!), and the conventions which restrict the amount of help a supervisor can give, apply. Nevertheless, most supervisors will be happy to read and to comment on drafts of sections of your project report before you hand it in, if you give them enough time to do so. It's also a good idea to report before you hand it in, if you give them enough time to do so. It's also a good idea to ask your supervisor to suggest some high quality past projects in a similar field to your, and to look them up in the departmental library. This will give you an idea of what is required.

SUMMARY

Good writing is difficult, but it is worth taking the trouble to write well.

In a summary write the main key points precisely in a small paragraph.

Unit–7

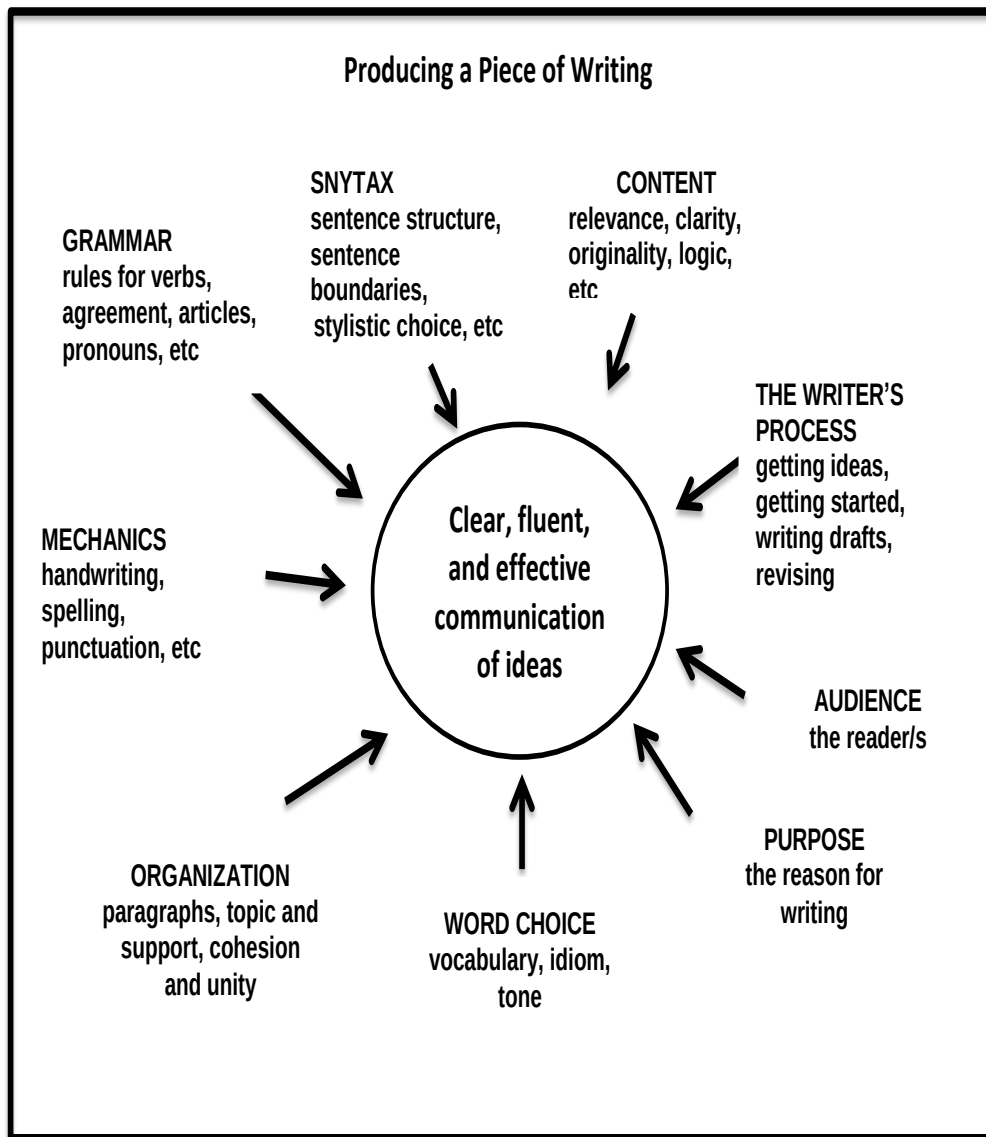
CREATING THE PROFESSIONAL REPORT

Written by: Mubadirah Durrani
Reviewed by: Syed Zafar Ilyas

OBJECTIVE

To present professional reports having command over mechanical and graphical elements.

7.1 MECHANICAL ELEMENTS OF PROFESSIONAL REPORT



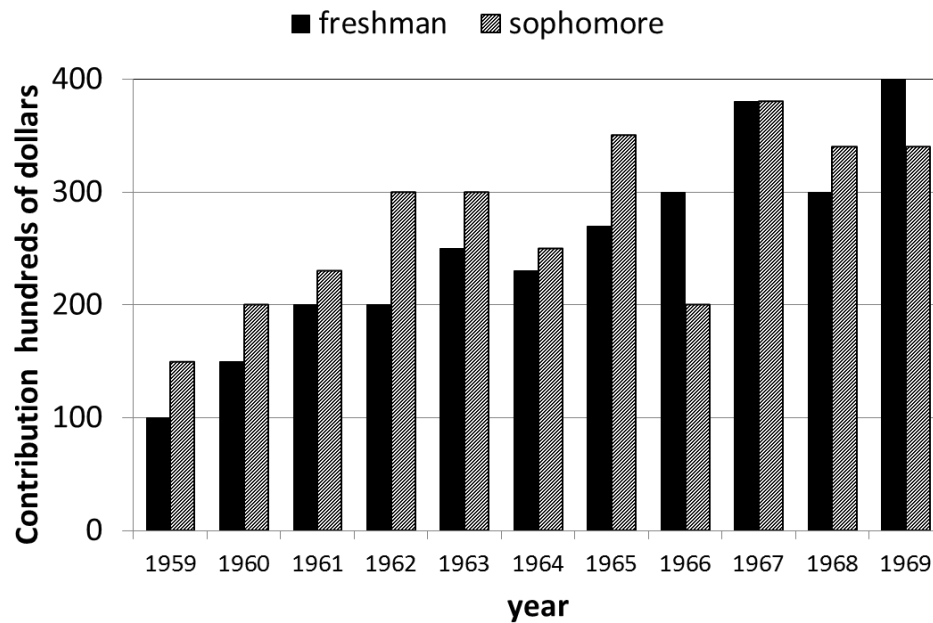
7.2 GRAPHICAL ELEMENTS

Graphs help to exhibit comparison of data in an easy way. It facilitates to explain in discussion part and makes assessment comprehensive. There are line graphs, bar graphs and Pie-charts so that one can use according to the nature of data.

BAR GRAPHS

A bar graph is similar to a line graph except that bars (extending from either the vertical or the horizontal axis) are used instead of dots and lines.

Freshman and sophomore class contributions to the Red Cross.



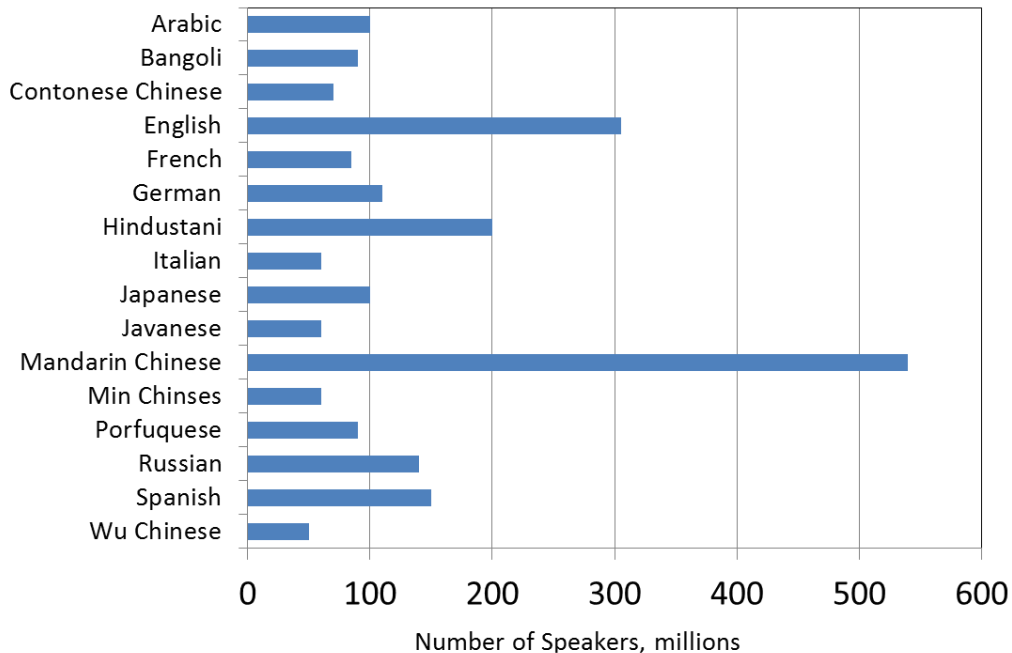
Exercise: Study the bar graph carefully. Then write information in a table.

Year							
Fresh man							
Sopho more							

EXERCISE

Study the following bars graph carefully. Then answer the questions.

Language of the world spoken natively by 50 million or more persons. (Based on data from the information. Please Almanac, 1969).



Answer the following questions by writing true or false on the line at the left.

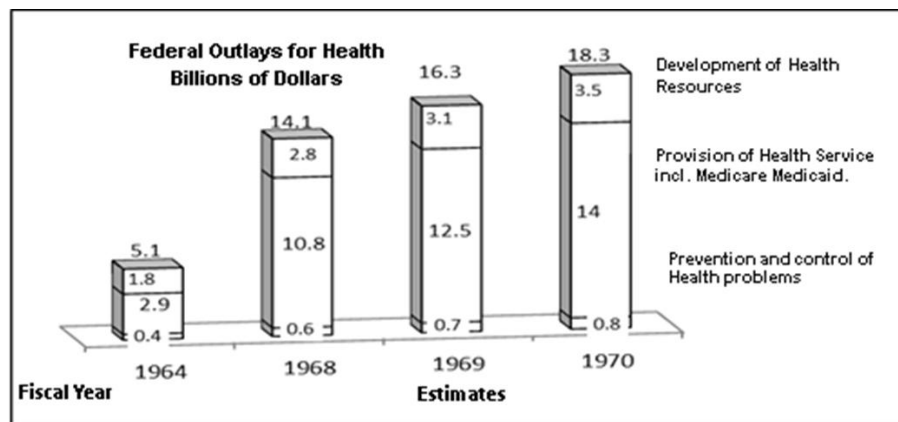
1. This table shows the number of speakers of all the languages of the world.
2. The number of speakers is shown on the vertical axis.
3. The number of speakers is indicated in millions; for example, 100 means 100,000,000.
4. There is only one language in the world with more than 300 million speakers.

5. There are twice as many speakers of Hindustani as there are of Japanese.

Answer the following questions by writing on the lines provided.

6. About how many native speakers of English are there in the world?
.....
7. Which dialect of Chinese has the most speakers?
.....
8. What language has as many native speakers as Arabic has?
.....
9. Which languages have half as many native speakers as Japanese?
.....
10. Which language ranks fifth in total number of native speakers?
.....

Exercise: Study the following bar graph carefully. Then answer the question below it.

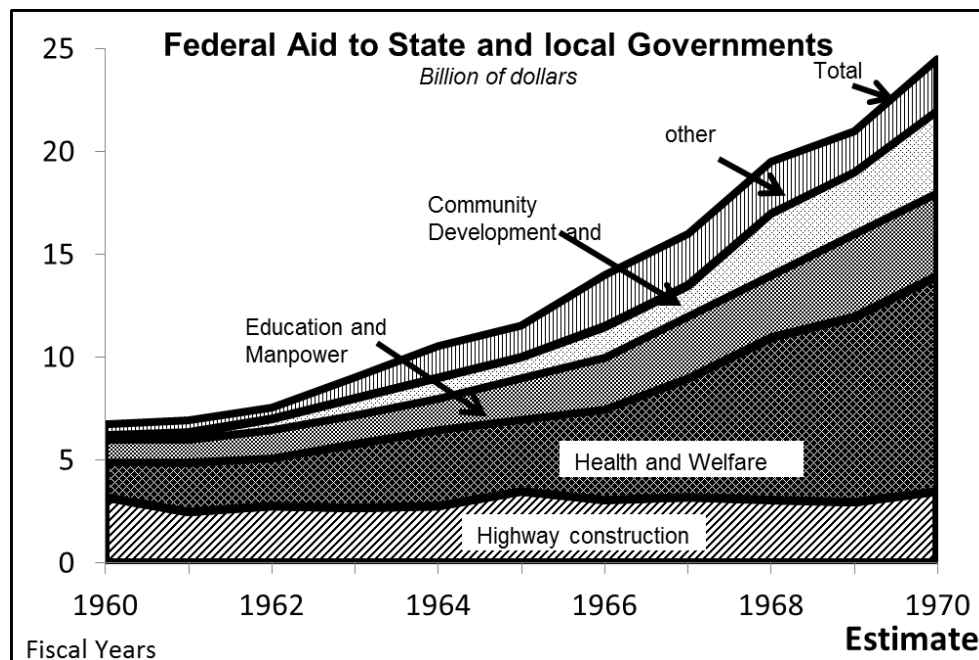


1. In 1967, what was the approximate total amount of federal aid to state and local governments?

.....

.....

Exercise: Study the following graph carefully. Then answer the questions on the opposite page.



1. List the five categories of federal aid to state and local governments that are shown in the graph.

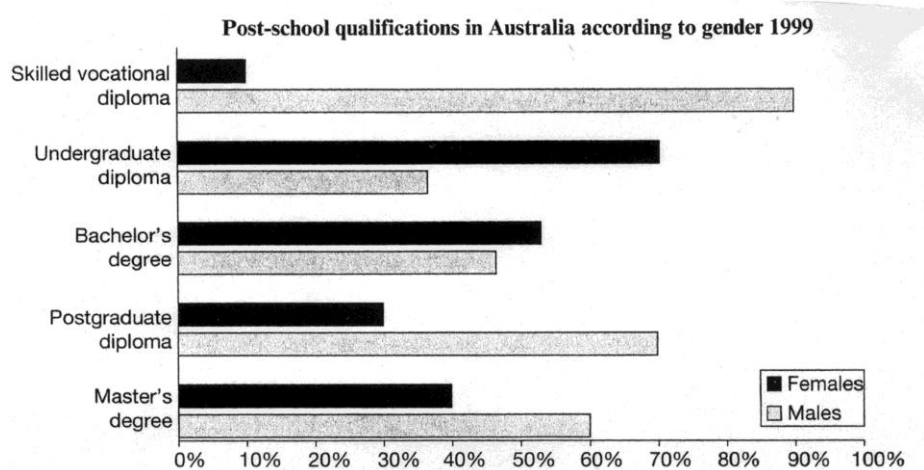
- a.
- b.
- c.
- d.
- e.

2. Of these five categories, which received the least federal aid in 1960?
.....
Which received the most federal aid in the same year?
.....
3. In the 1970 estimate, which category will receive the least federal aid?
.....
Which will receive the most federal aid?
.....
4. In the 1970 estimate, which category will receive about the same amount of federal aid as it did in 1960?
.....
5. In 1967, what was the approximate total amount of federal aid to state and local governments?
.....
6. Explain the graph in a paragraph. (150 words)

7.3 GRAPH EXPLANATION

The chart below shows the different levels of post-school qualification in Australia and the proportion of men and women who held them in 1999.

Summarise the information by selecting and reporting the main features, and make comparisons where relevant.



The chart gives information about post-school qualifications in terms of the different levels of further education reached by men and women in Australia in 1999.

We can see immediately that there were substantial differences in the proportion of men and women at different levels. The biggest gender difference is at the lowest post-school level, where 90% of those who held a skilled vocational diploma were men, compared with only 10% of women. By contrast, more women held undergraduate diplomas (70%) and marginally more women reached degree level (55%)

At the higher levels of education, men with postgraduate diplomas clearly outnumbered their female counterparts (70% and 30%, respectively), and also constituted 60% of Master's graduates.

Thus we can see that more men than women hold qualifications at the lower and higher levels of education, while more women reach undergraduate diploma level than men. The gender difference is smallest at the level of Bachelor's degree, however.

Unit–8

REPORTS

Written by: Mubadirah Durrani
Reviewed by: Syed Zafar Ilyas

OBJECTIVE

To write project reports research papers and electronic documents.

8.1 WRITING YOUR REPORT

Having read your book and recorded your impressions while they were fresh in your mind, you are ready to assemble your notes. You will have the factual material on hand, and your answers to the questions you asked yourself. These answers may be lengthy. They may emphasize one phase and not another.

Arrange your notes in basic outline form. Include all necessary information; omit details that add little to the picture. Sum up your general impression in the conclusion.

Now write your first draft. Check it thoroughly for correct grammar, usage, spelling and punctuation. Polish your first draft by writing in your corrections. Read it over and over. Does your enthusiasm or dislike for the book carry through to you as it should to the reader of your report? If your report plods along, try to introduce some color and emotions so that your reader will be tempted to investigate the book.

8.1.1 HOW TO WRITE A ONE-PARAGRAPH REVIEW

You may be asked to write only a few full-length reports in your previous classes. But you may be required to submit many one-paragraph reviews.

Offhand, a one-paragraph review sound like something you can toss off without much thought. In two or three sentences you can identify the book and say whether you liked it or not. But that is not enough.

Being able to write a closely packed summary paragraph will help you in many writing chores—science reports, examinations, business letters, committee reports.

When you have a concise review to write, keep in mind the answers to all of the check-list questions, but plan to use only some of the answers. The trick is to select the questions most appropriate for this particular form of writing. Eliminate all but the most important comments.

The success of a one-paragraph review depends on the compactness of your thoughts and your sentences. Every idea and every word must count. There is no

room for details, descriptions and dialogue. Your comments must be brief, positive, concrete and put together in simple direct sentences.

Arrange your notes in outline form. Study them coldly. Which details are vital to your report and which ones are less important? Can two important details be included in the same sentence? Try it. Can you express your feelings about the main theme in one sentence? Try it.

Write your first draft as compactly as you can. Go over it word by word to see if you can't condense it even further. Aim at leaving only the gist of the matters intact. Ruthlessly cut out the rest.

Make a clear, clean copy of what is left. Does it make one paragraph? Have you left out something really vital? If you put it back in, will it upset the whole paragraph? Try it.

When you are satisfied that you have written the tightest, yet clearest paragraph possible, make a final copy and hand it in.

8.2 HOW TO WRITE A SCIENCE PROJECT REPORT

Knowing how to write clearly and with authority is just as important for the scientist as it is for the journalist or the scholar. There are two reasons why you should learn how to write good reports of your science projects. First, by putting your work into words you will gain a better understanding of your project. Second, knowing you are going to write a report and what should be included in the report will help you plan and conduct the project. You will have to apply to writing the same scientific method – planned, orderly procedures – that you use in your science work.

The form and content of a science report will vary with the kind of project you are reporting on. Not all of the following items must be included in every report. For example, if a project did not involve an experiment, the report would not include the items about experiments.

8.2.1 THE TITLE

A good title will distinguish your project report from all others which have to do with the same general area of study. Titles of scientific reports should convey important information rather than show off the author's cleverness. Thus, the title, "Snakes Are Our Friends", would not be a suitable title for a research project on

snakes. It does not tell enough, specifically, about the contents of the report. A better title would be, “The Diet of the North American Grater Snake”. This title tells exactly what the report is about.

8.2.2 ABSTRACT OF THE REPORT

A 100 to 150 words summary of the report should follow the title. This summary should include in one paragraph a statement of the purpose of the project, general methods of procedures used, and principal findings and conclusions. The purpose of an abstracts is to give the reader enough information for him to decide whether or not to read the whole report.

8.2.3 BACKGROUND INFORMATION

This section of the report should give the reader enough information to be able to understand the history and the importance of your problem. It will often include a review of other people’s findings. If well written, this section will explain your motives for undertaking the project and will stimulate the reader. Background information may contain references to books and articles, indicated by footnotes. This section is often called simply “Introduction”.

8.2.4 THE PROBLEM

If your report is based on the study of a specific problem, the problem should be clearly stated or defined. You should tell whether you are searching for or testing hypotheses, suggesting a theory, or merely reporting some observations made under clearly specified conditions. While discussing the problem, you may often want to show its relationship to other problems. The best reports include a discussion of the relationship of the problem to existing theories.

8.2.5 HYPOTHESE TO BE TESTED

If one or a series of experiments is being reported, all hypotheses being tested should be stated. This section may also include a discussion of what the possible experimental results will mean in terms of accepting or rejecting the hypotheses.

8.2.6 PROCEDURE

Anyone who reads your report should, with the proper materials and facilities, be able to repeat your experiment or observations and obtain similar results. Therefore, you

must describe in detail all of the equipment and conditions (temperature, pressure, or any other environmental factors which could affect the outcome). Photographs and drawings can be used as well as words. Every step in your procedure should be carefully explained. If you discover that some method of doing something will not work, mention it so that others will not repeat your mistake.

8.2.7 DATA OR OBSERVATIONS

Numerical data such as measurements and other statistics are best presented in the form of tables. Observations i.e., descriptions of flowers or of the behavior of animals – should be reported in a simple, logical manner. Frequently such observations are recorded chronologically, in the form of a diary.

8.2.7 ANALYSIS AND INTERPRETATION OF DATA

In simple, short reports, this section may sometimes be included with the preceding one. The analysis of numerical data may include graphs and diagrams. Statistical methods may also be used to discover relationships. Nonmathematical observation should be analyzed and interpreted in terms of the hypotheses that were being tested. Put more simply, if the experimental data support or confirm your theory, you should organize them in the way that most clearly shows this support or confirmation.

8.2.9 CONCLUSIONS

In this section, each hypothesis should be re-examined and rejected if the data show it to be wrong. Hypotheses which are supported by the research can be tentatively accepted for further testing. You will seldom be able to fully accept a hypothesis. This section should also, when appropriate, discuss the effect of rejecting or accepting the hypotheses on the theories previously discussed.

8.3 IMPLICATIONS OR RECOMMENDATIONS

This section is sometimes devoted to “generalizations”. In this next-to-last section of your report, you should discuss any meaning your research may have for a better understanding of a broad area of science. You should also include a discussion of any new problems (or revisions of hypotheses) that have been suggested by your study. Never try to draw illogical connections between your research results and unrelated areas of interest and study. For example, it would be silly to say, “Some foolish people kill our valuable snakes,” in a report about “The Diet of the North American Garter Snake”.

8.3.1 SUMMARY

Always conclude a scientific report with a brief summary of the principal results of your investigation. It is often convenient and desirable to do this by means of a simple opening statement such as “The principal findings of the study were:” and then list the findings.

8.3.2 HINTS ON STYLE

When writing a scientific report, avoid the use of words that add no scientific understanding to what you are saying. Uncommon words and names should be defined. Use the third person as much as possible (that is, do not use the words “you” and “I”). When in doubt about a word, use a dictionary.

8.4 ORGANIZING YOUR NOTES

Once your notes are taken, how do you use them? This also depends on your subject. Some subjects suggest a chronological order, others a logical order. For instance, if you are doing a paper on Keats’s development as a poet you might first arrange your notes chronologically – notes on early poems, notes on middle poems, notes on late poems. But if your subject is an analysis of the themes of Keats’s poems, you might try to arrange your notes by themes, running various classifications until you have one that seems to make sense. Or you might find, sometimes that two levels of organization are necessary. For instance, certain themes of Keats’s poems might be characteristic of certain periods. Then having established one type of classification (by theme) you might run another type (by chronology). Notes are flexible. You can use them as a device to help your thinking as well as to help you organize your material.

Notes record questions and issue. The different authors you have consulted have had individual approaches to the general subject, different interests, and different conclusions. As you work over your cards you can locate these differences and try to see what they mean to you in your special project. Ask yourself if there is any pattern of disagreement among the authors you have consulted. List the disagreements. Are they disagreements of fact or of interpretation? Compare the evidence and reasoning offered by the authors who are in disagreement. Can you think of any new evidence or new line of reasoning on disputed points? Can you think of any significant points not discussed by your authors? What bearing would such points have on their conclusions? Again, use your notes as a device to help your thinking.

8.4.1 PLANNING YOUR PAPER

By working over your notes and thinking about ideas suggested in them you will probably strike on some vague general plan for your paper. But do not commit yourself to the first plan that comes into your head. Consider various possibilities. Then when you have struck on the most promising, try to work up an outline on that basis. You will undoubtedly, with the help of sentence outline, state exactly what you mean.

8.5 HOW TO WRITE A RESEARCH PAPER

The research paper draws its material from many sources. Its aim is assemble facts and ideas and by studying them to draw new conclusions as to fact or interpretations, or to present the material in the light of a new interest. The basic method should be the same to collect the facts and interpret them. The paper can be intelligent, well informed, interesting and original in the conclusion, but first of all your work should be systematic.

8.5.1 FINDING INFORMATION ON YOUR SUBJECT

The first step toward making your paper systematic is to learn how to investigate your subject. It is important to get original source of information for fact related to subject in the first place. Secondly you can use secondary sources in shape of books, repots, periodicals, encyclopedia or research papers. Material that is as close as possible to the original source can be utilized. If facts are not dependable, your work will gain no value.

8.5.2 YOUR WORKING BIBLIOGRAPHY

The working bibliography should be kept on cards (3" x 5") of same size with only one entry on card. It can be arranged easily in alphabetical order according to need. The entry on card contains all the basic information about a book or article: the author's name with the last name first, title of work, the volume number if any. The place and date of publication and name of publisher.

8.5.3 TAKING NOTES ON YOUR READING

You must take notes on your reading and write relevant material and if there is special point, you can also mention page number with it. It can help you put everything in order and help you writing with clarity and coherence.

8.6 ELECTRONIC DOCUMENTS

Be sure to include enough information so that your reader can understand what you are requesting. Provide a bit of background or context if necessary. State your request clearly.

Take extra steps to minimize the e-mail exchange, for example, if you are requesting an appointment, state your purpose and name the times that you could come in your initial message. Your respondent may then be able to answer you with only one additional message.

Quote selectively and briefly from any prior message to provide context and background. Although sometimes it's good to quote an entire exchange so as to keep a record of what's been said and decided, often that's unnecessary and simply ends up making a message too long and cluttering the screen.

Many professors advise that you **think about why you are sending an e-mail message**. Are you asking something that could easily be checked if you took a few extra steps yourself? For example, e-mailing a professor simply to ask when her office hours are can be annoying when the office hours have been clearly announced on the syllabus already. On the other hand, e-mailing for an appointment is just fine. Are you asking a question privately that might be better as on the course conference, where all the students might usefully see the response? Are you e-mailing to lodge a complaint or to ask for a letter or recommendation or to seek help with a problem set? In these cases, personal contact and an office visit might be much better.

Be respectful, and think about what kinds of things might sound odd or offensive to your professor. For example, don't say flippantly that you slept through that professor's class, or talk about your love life, or bash chemistry or math or writing.

Wellesley faculty think very highly of their students, and you will soon find that they will work closely with you and that you will feel quite comfortable entering an intellectual relationship with your professors. E-mail has often been seen as a democratizing system of communication that flattens hierarchies and that allows people of all sorts to communicate comfortably and freely with each other. Everyone values the ease of communication that e-mail offers us. Most professors are overwhelmed by the vast number of e-mail messages they receive; some are shocked by the sheer effrontery of some of those messages; many become frustrated if e-mail consumes so much of their time and emotional energy that

they cannot then spend valuable personal time with you. We hope these tips will help you – and your professors – conserve some of that emotional energy.

And as for that emotional energy, here's one last tip:

Be kind to yourself if you make an electronic faux pas. All of us – even those who have studied electronic communication for years – have made some pretty monstrous errors. We learn from our mistakes, and we learn to forgive ourselves (and others).

WCPSC Faculty Director: Wini Wood

Maintained by: Anne Manning

Date Created: 25 August 2008

Generalized from an email to a UCSC professor

Subject: Possible undergraduate research opportunities

Dear Dr. Professor,

I am a **(year, major)** at **(university)** and I am writing to ask about opportunities for undergraduate research in your lab beginning **(time period)**. I have conducted undergraduate research on **(topic)** with **(names)** in **(program or class)**. **(Expression of interest in the topic)**. I would like to continue a path or research on **(topic)** and would ultimately allow me to **(career goal)**. I am especially interested in your previous work on **(describe a paper or talk)**.

I have attached my CV and unofficial transcript to this e-mail, but if there is additional information that I have not included that you would like, I would be happy to provide it to you. Thank you for your consideration.

Sincerely,

Ambitious Student

Email address.

From University of Virginia, How to Successfully E-mail Professors

Dear Dr. Smith,

My name is David Wu and I am second year biology major at UVa. In my introductory and upper-level coursework, I have developed a passion for science and am extremely interested in pursuing independent research as an undergraduate. An extensive research experience will greatly help me consolidate my future career choice.

I am personally greatly interested in the molecular biology of stem cells. Recently I read your 2011 paper on the role of microRNAs in the differentiation of muscle stem cells and became fascinated by your work. In particular, I found it amazing that microRNAs can alter the fate of a cell in such profound way. If possible, I would love to start working on a long-term project in your lab beginning this summer.

Would you be available to meet sometime this week to discuss your research? I would also be happy to volunteer in your lab for a few weeks before we commit to anything to see if this is a good match. My transcript and resume are attached in case you are interested. I look forward to hearing from you!

Thank you,

David Wu

Unit–9

PROGRESS REPORT

Written by: Mubadirah Durrani
Reviewed by: Syed Zafar Ilyas

OBJECTIVE

To produce various proposals, reports and articles for projects and conferences.

9.1 GUIDELINES FOR WRITING A PROGRESS REPORT

People write progress reports to keep interested parties informed about what has been done on a project and about what remains to be done. Often the reader is the writer's supervisor. As a result the tone should be serious and respectful. Even though progress reports are often in the form of a memo, the writer should be careful to write formal, standard prose. Progress reports represent not only the writer's work but the writer's organizational and communication skills.

Progress reports can be structured in several ways. The following suggested pattern helps the writer cover essential material.

9.2 HEADING

If the progress report is a memo, it should contain the following standard elements:

- Date: Date the memo is sent
- To: Name and position of the reader
- From: Name and position of the writer
- Subject: A clear phrase that focuses the reader's attention on the subject of the memo.

9.3 PURPOSE STATEMENT

Because the reader is busy, get right to the point. Imagine you are meeting the reader in the hall, and you say, "I wanted to talk to you about this." Use the same strategy for the first line of the memo's body. Try saying out loud, "I wanted to tell you that" and then start writing whatever comes after that prompt. Often such a sentence will begin something like this." "Progress on setting up the new program in testing is going very well." If there is a request somewhere in the memo, make it explicitly up front; otherwise, your reader may miss it.

9.4 BACKGROUND

Usually in the same paragraph as the purpose statement, the writer gives the reader some background information. If the occasion demands a written progress report instead of a quick oral report, it is probably the case that the reader needs to be reminded of the details. Tell the reader what the project is and clarify its purpose and time scale. If there have been earlier progress reports, you might make a brief reference to them.

9.5 WORK COMPLETED

The next section of a progress report explains what work has been done during the reporting period. Specify the dates of the reporting period and use active voice verbs to give the impression that you or you and your team has been busy. You might arrange this section chronologically (following the actual sequence of the tasks being completed), or you might divide this section into subparts of the larger project and report on each subpart in sequence. Whatever pattern you use, be consistent.

9.6 PROBLEMS

If the reader is likely to be interested in the glitches you have encountered along the way, mention the problems you have encountered and explain how you have solved them. If there are problems you have not yet been able to solve, explain your strategy for solving them and tell the reader when you think you will have them solved.

9.7 WORK SCHEDULED

Specify the dates of the next segment of time in the project and line out a schedule of the work you expect to get accomplished during the period. It is often a good idea to arrange this section by dates which stand for deadlines. To finish the progress report, you might add a sentence evaluating your progress thus far.

9.8 PROPOSALS

Science or Engineering Proposals

Goals and Objectives:

State hypotheses or theoretical predictions that you will test. If your project is part of a larger collaborative research agenda within your mentor's lab or group, describe both the overarching research goal and the particular component(s) that you are focusing on. What are your specific aims for this particular phase of the project?

Significance:

Critique the most relevant recent peer-reviewed papers that set the stage for your hypotheses. Are you challenging or refining current knowledge of the mechanisms, technologies, or phenomena of interest? Demonstrate your grasp of what has been done on this topic or in related areas, and show how your research builds on and contributes to this body of knowledge.

Project Plan:

What are the key steps in the experiment or modeling effort that will test your hypothesis? What quantitative analyses will enable you to interpret your observations or refine your model? Spell out your specific role and responsibilities, and your intellectual ownership of these responsibilities. Provide a logistical plan including your timeline and milestones for progress.

9.8.1 SOCIAL SCIENCE PROPOSALS

Proposals in the social science may share some of the features of either science or humanities proposals. For example, an econometric comparison of tax policy effects would need to discuss a statistical framework, whereas an ethnographic account of folk stories would outline a very different kind of qualitative method.

Goals and Objectives:

State the questions or hypotheses you will examine, making clear the level of focus and the quantitative or qualitative nature of your goals.

Significance:

Provide a literature review that connects your research questions to current thinking in your field. Select and critique recent works that provide a foundation to the way you are stating your research questions, and that show your knowledge of what other scholars are already working on.

Project Plan:

Provide a detailed methodology that will enable you to achieve your project goals. What observations datasets, or transcripts will you need to answer your question? What analytical framework (statistical or qualitative, as appropriate) will let you draw sound conclusions? Provide a logistical plan, including your timeline and milestones for progress.

9.8.2 SPECIFIC PROPOSAL GUIDELINES FOR ARTS OR DESIGN PROJECTS

Goals and Objectives:

What are the objectives of your project? Give your most focused vision for the finished project or production. Describe any key expressions, impressions, or insights that you intend, and the audience that you intend them for.

Significance:

Cite sources of artistic/design inspiration for the project. What is the relationship between your proposed project and past or current work by other artists/designers? In light of the current dialog across related media, how is your vision unique, interesting or original?

Project Plan:

What techniques will you use? Why and how will you use them? What area or media will you be working in? What is your conceptual approach to this project? What concerns do you address? How do you envision your final product? For productions or exhibitions, provide a detailed timeline with the projected start date, including: pre-production research; production schedule, itemizing tasks and allocating time and delegation for each; post-production plans, if applicable. How will the final work or production you envision enhance your interests, skills, opportunities for further work? How will your plans for presenting this work affect your audience?

Preparation:

What formal or informal training and mentoring have you had, or will you need? How do your past experience inform this project? Link to online samples of your previous work if possible.

9.9 SPECIFIC PROPOSAL GUIDELINES FOR SENIOR SYNTHESIS PROJECTS

Goals and Objectives:

As concisely as possible, what do you hope to accomplish? Be clear about your aims for this phase of your project, and how those aims will help further your broader intellectual aspirations.

Significance:

Why have you chosen this intellectual goal? Describe the different intellectual aspects of your undergraduate academic and extra-curricular experience and how your proposed project will allow you to integrate these components. Fundable proposals will reflect on the first eight quarters (or more) of your undergraduate work to illustrate what inspires and informs the way you are integrating these ideas. What is the relationship between your project idea and what others have done or discussed (a literature review may also be relevant)? Why will your mentors, advisors, teachers and peers be interested in what you are doing?

Project Plan:

What will you do? What are the steps that will allow you to complete the synthesis and reflection you have described above? When will each of these steps be completed, and how will you gauge your progress? How will you reflect on your project experience and share those reflections with others? What form will your final product take and how will it be shared with the area community? Will you earn academic credit for any part of your project?

Preparation and Mentoring:

What skills (from the classroom or from somewhere else) will you bring to bear on this project? What courses provided you with the most important intellectual background? Do you have hands-on experience that will be valuable? Why are your mentors and their expertise a good fit for this project? How have they helped you so far, and how will they continue to mentor you as your project unfolds?

9.10 PROPOSAL GUIDELINES FOR CONFERENCE GRANTS

Conference Grants support your presentation at a scholarly conference or meeting and UAR (Undergraduate Advising and Research) requires a different type of proposal for this application. Your proposal should include the sections outlined below.

Proposal title: Provide the title of your conference presentation.

Conference Abstract:

Provide the abstract of the paper or poster that has been accepted for presentation. Usually, this abstract describes your project's objectives, significance, methods, and conclusions. If you think additional background or explanation on this project would be helpful, include this information succinctly.

Your Role in the Presentation:

Conference Grants are for presentations that you are leading. If you have co-authors, indicate the extent of your role in presenting and responding to questions from the audience.

The Meeting or Conference that you are attending:

Include the name of the conference, information on the society or association sponsoring it, with dates and location, as well as details of the session at which you are approved to present. Include the abstract acceptance notification you received from the organizers (your name and presentation title should be identified in that official correspondence).

The process whereby your abstract was accepted: Conference grants are for competitive peer-reviewed selection processes.

Goals of Attending this Conference:

Describe specific sessions or meetings that you will attend, beyond the one at which you will present. Describe what you would discuss with scholars, what you hope to learn at this conference, which you will not be able to do if you did not participate in this conference. How will these sessions and conversations fit with the other work you have done in and outside of the classroom at university?

9.10.1 GENERAL GUIDELINES FOR THE GRANT PROPOSALS

Your original project proposal is the core of your grant application. A good proposal will describe what you hope to accomplish, why those objectives are important, and how you intend to achieve them. The ideal format and language for your proposal will vary with your specific project, and with the discipline in which your project is situated. All good proposals will address the following questions concisely, in a manner clear to both specialists and non-specialists.

Goals and Objectives: What precisely are you hoping to accomplish?

Significance:

Why are your goals important? What kinds of previous work (others' or your own) do you use to justify your goals and objectives? Who will be most interested in what you do?

Project Plan:

How will you accomplish these goals? How does your project design lead to the outcomes you envision (logically and logistically)? Will experts in your discipline agree that your plan is a good one? What is your timeline? What final product(s) will result from your plan?

Resources:

How will you draw on your mentors' expertise? Do you have letters of support from contacts (at field sites or other institutions) who will be critical to your project's success? Are you seeking or do you have any other sources of funding? Are there additional sources of information, suppliers, or equipment that you will rely on?

Preparation:

What specific steps have you taken to prepare for this project? Where did you get the skills you will need? What other project requirements have you completed (e.g. human subjects approval, travel and safety plans)?

Budget:

How much money do you need, and what will it be used for? How does each line-item in your budget support a step in your project plan?